Dual Forests and Homological Progress Certificates
for Compressed Unknot Hierarchies

Research contribution prepared for the ProveIt project

12 October 2026

Source baseline: `8cb7d9d25ac8cb64a9d147304e3441e98e94c7a4`

Abstract

We develop two concrete contributions toward a compressed implementation of Lackenby's unknot-recognition programme. First, for the authenticated bounded core of any normal cut in the current finite-source contract, we reconstruct the two sides of every source endpoint-gap two-handle and prove that their dual multigraph is connected to exposed boundary. Every rooted spanning tree therefore cancels all source three-handles, relative to the existing bundle ports, with a trace of size independent of normal-coordinate magnitudes. A maximum-weight tree minimizes retained additive attaching incidence. An explicit five-tetrahedron solid torus proves a strict advantage: one forest leaves a persistent stopping set, while the weighted forest permits a complete geometric certificate. We also characterize successful forests by stopping-set transversals, derive a graphic-rank obstruction, and refute a general matroid exchange hypothesis for raw-peelable rows.

Second, we give an integer-certificate alternative for membership in the rational image of bundle homology. Under explicitly authenticated hierarchy hypotheses, an outside-image certificate forces a strict decrease of Lackenby's quadratic potential, with an exact raw-cut defect formula. The algebraic checker is implemented separately; its geometric compiler remains open. The integrated top reduction increases certified classifications from 85 to 92 on the pinned 106-case cut corpus, preserves every prior success, and has independent replay and explicit-cut validation. These are proved local advances. We identify the missing global depth, branching, pattern-transport, and encoding estimates needed for either announced quasi-polynomial bound.

Contents

1 Research objective, source baseline, and quantitative target **2**

2 Eliminating top handles after a compressed normal cut **4**

3 Relative bundle quotients and certificates of strict progress **15**

4 Source-authenticated implementation and integration **23**

5 Reproducible validation and performance measurements **25**

6 From local improvements to a global complexity theorem **29**

7 Further research questions and concrete next investigations **32**

A Artifact contents and reproduction **35**

B Claim ledger **37**

1 Research objective, source baseline, and quantitative target

The immediate objective is to turn compressed normal-surface data into more useful, independently checked geometric operations. The eventual objective is a deterministic unknot recognizer with a proved quasi-polynomial worst-case bound in the number of crossings of its input diagram. These objectives are connected, but they require different theorems. A fast orbit routine, a successful terminal handlebody test, or a polynomial-size positive certificate does not by itself control the total number of surfaces or hierarchy states an algorithm must inspect.

This contribution supplies a new source-relative compiler step and a separate, exact certificate of a sufficient condition for hierarchy progress. The first is integrated into the existing cut-classification pipeline. The second is an algebraic research component whose future geometric interface is specified explicitly. Their proofs, implementation, negative examples, and experimental limits are part of one deliverable.

1.1 The pinned state of ProveIt

All baseline comparisons use repository revision

`8cb7d9d25ac8cb64a9d147304e3441e98e94c7a4`.

The snapshot is from 12 October 2026. The 905 original runtime-directory blobs were authenticated against their Git blob identifiers. The delivered patch is relative to that revision, rather than an unpinned moving branch. The incoming research packages in the same snapshot were also inspected before choosing the present contribution [12, 8, 10, 7, 11, 9].

The relevant existing pipeline starts from a compact, connected, orientable finite triangulation with one torus boundary and an admissible normal coordinate vector. It constructs the actual cut in compressed form: a bounded complementary handle core, interval-bundle families with binary multiplicities, and authenticated ordered interfaces. The terminal operator then certifies a census of handlebody components when its geometric cancellation and bundle-attachment criteria apply.

The word `COMPLETE` in that operator means an exact classification of the *supplied cut*: every component is certified to be a handlebody and its genus and multiplicity are computed. It is not a complete decision procedure for arbitrary handlebodies, and it is not a standalone knot verdict. A nontrivial knot exterior can have a cut consisting of handlebodies. The word `INCONCLUSIVE` records the failure of this particular positive compiler, not a proof that a component is a non-handlebody or that a knot is nontrivial.

The pinned terminal operator certifies 85 of the maintained 106 source cuts. Ten unresolved inputs are rejected immediately because the source core contains three-handles. The first theorem family below removes that entire *structural refusal*: every source three-handle can be cancelled after the normal cut. Later positive cancellation still need not complete. On this corpus seven of the ten inputs become certified, giving 92 successes.

1.2 Prior results that this article uses

The following distinctions prevent repeated work from being presented as a new discovery.

- The incoming *Genus Budgets and Native Interfaces* contribution already proves completeness of the ordinary source two/three queue before a normal cut and with the initial empty boundary pattern. Classical two/three cancellation is also prior work. The new result concerns the *compressed complementary core after an arbitrary normal cut*, its endpoint-gap side table, preservation of bundle ports, and independent source replay.

- The unsigned stopping-set characterization of positive raw one/two cancellation, complete column peeling, and the associated unit-permanent minor criterion are already in the incoming stopping-core work. Our new use couples those obstructions to the set of allowable source dual forests and realizes a strict forest-choice advantage in an actual small manifold.
- Saturated bundle cancellation is already incorporated in the pinned baseline. Consequently, 78 successes is an obsolete comparison point for this patch. The appropriate baseline is 85.
- The chain-space separation theorem used for the second component is finite-dimensional rational linear algebra. The contribution is its exact certificate contract, independent integer replay without trusting a purported complete bundle-cycle basis, and its precise use with Lackenby's equality case. We do not claim a new general linear-algebra duality theorem.

These results extend the repository's current research programme. Assertions of priority beyond that programme would require a broader historical review; none is needed for the geometric and algorithmic claims proved here.

1.3 What exactly has Lackenby announced?

The available primary sources contain two quantitative formulations. The February 2021 talk displays a bound of the form

$$n^{O(\log n)} = \exp(O((\log n)^2)),$$

and discusses logarithmic hierarchy depth. The March 2021 Oxford handout instead displays

$$2^{O((\log n)^3)} = n^{O((\log n)^2)},$$

with a corresponding squared-logarithmic depth formulation. Here n is the crossing number of the supplied diagram. Both bounds are quasi-polynomial, but they are quantitatively different; the February bound is the stronger target [2, 3]. The Oxford announcement also describes an $n^{c \log n}$ bound [6].

The July 2026 preprint *Incompressible surfaces, hierarchies and unknot recognition*, already included among the project's source material, develops the geometric hierarchy machinery [4]. Its Section 9 explicitly leaves running-time estimates outside the precision of the stated algorithm. In particular, a quantitative iteration bound and a linear bound in quadratic handle complexity are not themselves either 2021 asymptotic claim. This article uses the preprint's stated theorems, while treating the global quasi-polynomial estimate as a research target requiring additional proofs.

1.4 Parameters and levels of certification

We distinguish four sizes throughout the article:

$$\begin{aligned} n &= \text{diagram crossings}, & t &= \text{source tetrahedra}, \\ L &= \text{maximum coordinate bit length}, & W &= \sum_i x_i. \end{aligned}$$

The source input has size $O(t \log(t+1) + tL)$ in a direct encoding. Normal weight W may be exponential or larger relative to t; a compressed operation must charge arithmetic to L, not loop once per normal disc. Later chain-matrix results use their explicit ranks and entry bit lengths as separate input parameters. There is no established identification of all those ranks with a polynomial in n along an arbitrary future hierarchy.

Three logical levels must also remain distinct. A chart-level verifier authenticates geometric incidence from the triangulation and coordinates. A local certificate, such as a dual forest, proves a transformation of that authenticated geometry. A global recognition theorem must additionally establish that the algorithm finds an adequate sequence of such operations, terminates correctly on both answers, and spends the claimed total time. The delivered native patch implements the first two levels for a new useful transformation. The final sections specify the missing work at the third.

2 Eliminating top handles after a compressed normal cut

The ordinary, uncut source presentation already has a complete two/three handle reduction. The incoming *Genus Budgets and Native Interfaces* article, Section "The complete initial empty-pattern compiler", states its queue rule and proves completeness by connectivity of the source one-skeleton. That result, and classical handle cancellation itself, are prior work [8]. The extension here concerns the bounded complementary core *after an arbitrary normal cut.* It identifies both sides of every endpoint-gap two-handle, proves that the resulting dual graph is still rooted-connected, preserves the authenticated interfaces with the parallelity bundle, and makes the reduction independently replayable without expanding normal coordinates. In particular, the existing `core_has_three_handles` refusal is not an unavoidable obstruction for this source family.

2.1 The source contract and the geometric side table

Let $\mathcal{T}$ be a finite triangulation of a compact connected three-manifold M with nonempty boundary. In the implemented contract, M is orientable and has one torus boundary; the topological argument below needs only the indicated finite-manifold and boundary hypotheses. Vertex links must be discs or spheres, and reversed self-identifications of an edge are excluded. Let S be an embedded normal surface, possibly disconnected, one-sided, or empty, given by admissible nonnegative integer normal coordinates x. By cutting along S we mean removing the interior of a small regular neighbourhood and taking the resulting compact manifold with rounded corners.

Use the authenticated source parallelity decomposition

$$M \setminus\setminus S = K \cup_A B.$$

Here B is the union of the specified interval bundles, K is the bounded complementary handle core, and $A = K \cap B$ is its port subsurface. This notation does not assume that B is globally a product. The source construction puts all its core-facing ports in boundary charts of the zero- and one-handles of K; the two-handles considered below are the *endpoint gaps* at source edges. The endpoint gaps are distinct from the interior edge gaps that belong to the parallelity bundle.

For a source edge e, choose its globally consistent orientation and let $v_-(e), v_+(e)$ be its two endpoint occurrences. Their global vertex identifiers may coincide. Put

$$w(e) = |S \cap e|.$$

Let V_{int} be the set of interior source vertices and use a formal symbol $*$ for exposed boundary. Define

$$\rho(v) = \begin{cases} v, & v \in V_{\mathrm{int}}, \\ *, & v \in \partial M. \end{cases}$$

Each $v \in V_{\mathrm{int}}$ supplies one source three-handle, irrespective of the normal coordinates. A source boundary edge supplies no core two-handle. An interior source edge supplies one endpoint-gap two-handle when $w(e) = 0$ and two when $w(e) > 0$.

Lemma 2.1 (The exact dual side table)**.** *The two belt-disc sides of the source core two-handles have the following owners:*

normal edge weight	*source gap rank*	*ordered side owners*
$w(e) = 0$	0	$(\rho(v_-(e)), \rho(v_+(e)))$
$w(e) > 0$	0	$(\rho(v_-(e)), *)$
$w(e) > 0$	$w(e)$	$(*, \rho(v_+(e)))$.

An occurrence of $*$ means an exposed face on the original boundary or on the boundary created by cutting S. Repeated interior owners are retained with their multiplicity: (v, v) means two intersections with the attaching sphere of the same three-handle, not one.

Proof. Give an edge neighbourhood its source product chart $D^2 \times e$. The normal surface meets the edge in $w(e)$ distinct points. Its complementary intervals are the gaps between successive normal points and the two endpoint gaps. A core two-handle is $D^2 \times J$, where J is an endpoint gap; its belt sphere consists of the two points at the centres of $D^2 \times \partial J$.

If $w(e) = 0$, the single gap runs between the two source vertices. A finite interior vertex contributes its vertex-link three-ball, while a boundary vertex contributes an exposed boundary chart. This gives the first row. If $w(e) > 0$, the initial gap runs from the initial vertex to the first normal point, and the final gap runs from the last normal point to the final vertex. Their normal-point sides are exposed after cutting. The intervening gaps, if present, lie in B and are not endpoint-gap core handles. This gives the second and third rows.

The argument is local at the edge. It does not require a coorientation of S: the two sides of each individual intersection point have local product neighbourhoods even when the whole surface is one-sided. Likewise, if the endpoint occurrences represent the same global vertex, they are still distinct ends of the edge chart and give distinct belt points. This proves the multiplicity assertion. □

There is a direct formula for reconstructing this table from the existing records. A zero-handle flag in a two-handle attaching cycle gives a local source edge $(t, [a, b])$, with $a < b$. The signed source edge equivalence relation gives a bit $\epsilon(t, a, b)$. Its ordered endpoint flags are $(t, a), (t, b)$ when $\epsilon = 0$ and $(t, b), (t, a)$ when $\epsilon = 1$. The three-handle vertex-link triangles assign an interior owner to each such vertex flag; unassigned flags must be authenticated boundary vertices. The edge weight can also be checked locally:

$$w(e) = x_{t,a} + x_{t,b} + \sum_{j=0}^{2} x_{t,4+j} - x_{t,4+q(a,b)}, \tag{1}$$

where $q(a, b)$ is the quadrilateral type that misses the edge $[a, b]$. Every term actually contributing to the right side is nonnegative. Normal matching makes this value the same around the entire source edge. An independent checker can therefore inspect *every* source zero-handle flag in the cycle and require the same oriented side pair; it need not trust a producer's chosen representative.

2.2 A universal rooted forest in the cut core

Define a multigraph G_x with vertex set $V_{\text{int}} \cup \{*\}$ and one edge for each source endpoint-gap two-handle, joining the owners in Lemma 2.1. Both parallel edges and loops are retained. Several boundary components of the cut manifold may be represented by the same formal root. This is an indexing device: the algorithm never glues these boundary components together.

Lemma 2.2 (Rooted connectivity survives normal cutting)**.** *Every vertex of G_x is connected to $*$.*

Proof. The source one-skeleton is connected. Fix an interior vertex v and choose an edge path from v to a source boundary vertex, stopping at the first boundary vertex encountered. Every edge before this final endpoint is an interior edge, since a boundary edge has both endpoints on the boundary.

If all edges on the path have weight zero, the first row of Lemma 2.1 turns it into a path in G_x ending at $*$. Otherwise, follow the initial weight-zero part of the path. At the first positive-weight edge, the endpoint gap incident to the current vertex joins that vertex directly to $*$. Either case gives the required rooted path. No upper bound on a nonzero edge weight has been used. □

Lemma 2.3 (Relative top-handle cancellation)**.** *Suppose an endpoint-gap two-handle has exactly one side on a surviving three-handle and its other side is exposed. This pair can be cancelled geometrically. The resulting presentation retains the original zero- and one-handle records and the original attaching curves of every unselected two-handle. The induced homeomorphism can be chosen to fix the port subsurface A.*

Proof. The belt sphere of a two-handle $D^2 \times I$ is the two-point set $\{0\} \times \partial I$. By hypothesis, the attaching sphere of the chosen three-handle meets precisely one of those points and does so once in its source product chart. It does not meet the other point, which lies on exposed boundary. This is the geometric two/three cancellation condition, with actual belt intersections rather than a signed cellular coefficient.

The elementary cancellation pushes the three-ball through this single two-handle to the exposed side. Equivalently, in the reversed handle picture it absorbs a zero-handle and its single connecting one-handle into a boundary collar. There is no other surviving three-handle using the selected two-handle, so no other top attaching sphere needs to slide over this pair. All remaining two-handles retain their attaching maps to the zero/one stage.

The source port rectangles come from interior bundle gaps of ranks $1, \dots, w-1$, whereas selected core gaps have ranks 0 or w. Source vertex three-ball neighbourhoods contain no bundle. Thus the source port rectangles lie in the zero/one boundary charts and are disjoint from the attaching regions and belt faces of the endpoint-gap two-handle and from the vertex-link attaching sphere. Shrinking the source collars slightly gives a cancellation neighbourhood disjoint from A. The cancellation is the identity outside that neighbourhood, so its induced pair-homeomorphism fixes A. This assertion concerns the presentation and the specified port subsurface. It does not assert a pointwise identification of the entire old zero/one submanifold with the entire new one. □

Theorem 2.4 (Complete cut-relative elimination of three-handles)**.** *For every authenticated source normal cut above and every spanning tree D of G_x, there is a geometric two/three cancellation sequence that deletes exactly the two-handles indexed by D and all source three-handles. The resulting core K_D satisfies*

$$(K_D, A) \cong (K, A), \qquad K_D \cup_A B \cong M \setminus\setminus S.$$

Its zero/one records and all unselected two-handle attaching cycles are unchanged. In particular, it has the same Euler characteristic as K.

Proof. Root D at $*$ and process its other vertices in any order in which parents precede children. For a vertex v, select the two-handle of its parent edge. Its v side still meets the attaching sphere of the three-handle v exactly once. The other side is either originally exposed or belongs to a parent three-handle already absorbed into the exposed boundary. It is therefore exposed at this stage. Apply Lemma 2.3.

Induction deletes one distinct two-handle and one distinct three-handle per nonroot vertex. No selected edge is a loop. A spanning tree has exactly $|V_{\mathrm{int}}|$ edges, so all three-handles are

deleted. Each step is relative to A, hence so is the composite homeomorphism. Extend this homeomorphism by the identity on B to recover the whole cut manifold. Each cancellation removes one handle contributing $+1$ and one contributing -1 to Euler characteristic, which proves the last assertion. □

There is also a direct embedded-ball interpretation of the proof. Remove the formal root from the selected tree. Every remaining tree component has exactly one selected half-edge leading to exposed boundary; two such half-edges would make a cycle through the root. Its source vertex three-balls, joined by the selected endpoint-gap tubes, form a regular neighbourhood of a tree and hence a three-ball. This ball meets the boundary of K in the single exposed cap disc. Its remaining frontier is also a disc, so removing it reverses the elementary attachment of a ball along a boundary disc. The source port charts miss these selected handles. Thus the ball removal is relative to A. This description does not replace the port contract by a homology computation: the tree and its product charts give an actual embedded ball.

The theorem covers an empty normal surface, a one-sided normal surface, and arbitrary binary multiplicities. It assumes the validity of the authenticated source charts; arbitrary three-handle summaries or group presentations are not substitutes. It does not assert that the resulting core is a handlebody. Raw one/two cancellation, bundle attachment tests, closed bundle bases, and later hierarchy conditions remain separate.

Corollary 2.5 (Compatibility with bundle consumers)**.** *Any subsequent construction whose geometric inputs are the original zero/one core charts, the unselected two-handle cycles, and the authenticated port subsurface can instead use K_D. In particular, source disk-bundle absorption and the existing relative bundle attachment constructions are valid after the top reduction. An absorbed additional two-handle is not an eligible edge of G_x and cannot be selected by the top-cancellation certificate.*

Proof. The required charts and attaching subsurface are preserved by Theorem 2.4. Thus the construction attaches exactly the same geometric pieces along the same port data. Added two-handles arise from the specified later attachments, rather than from source edge endpoint gaps; they have no source vertex-link side authentication and are kept outside the eligible graph. □

An arbitrary later boundary pattern need not be supported on A. It may cross an exposed face of a selected endpoint-gap two-handle. Transporting such a pattern is not supplied by the preceding theorem. The relative scope is precisely the authenticated bundle ports and any marks supported there.

2.3 Support dependence and small reusable certificates

Proposition 2.6 (The dual graph depends only on zero support)**.** *For a fixed source triangulation, G_x depends only on the set*

$$Z(x) = \{e : w_x(e) = 0\}.$$

Here an edge address is $(e, 0)$ when its weight is zero and $(e, -)$ or $(e, +)$ for its initial or final gap when its weight is positive. If $Z(x) = Z(y)$, these addresses identify G_x with G_y. Every rooted forest trace for one is therefore a valid trace for the other after independent source authentication. In particular, this applies to $y = mx$ for every positive integer m.

Proof. The interior vertices, boundary vertices, and oriented endpoint owners are fixed by the source. The side table has only the dichotomy $w = 0$ or $w > 0$; no positive magnitude appears in any owner. Hence the stated addresses give identical multigraphs. A rooted trace only records those edges and their endpoint owners, so the same root-before-child argument replays it. Finally, $w_{mx}(e) = mw_x(e)$ has the same zero set for $m > 0$. □

The normal source coordinates and any other source certificate still need to be checked for each input. The proposition reuses the combinatorial top trace, not the truth of source authenticity. An edge-zero bit vector of length $O(t)$ is a sufficient cache key together with the source triangulation identifier for a graph-only choice such as breadth-first search. A maximum-weight strategy may depend on finer data through its row costs: an already selected trace still replays at the same zero support, but need not remain cost-optimal there. Positive scalar multiplication preserves the full bounded-core incidence combinatorics and the fixed primal forest, so the raw-incidence costs, as well as a deterministically tied maximum-weight choice, are also unchanged under $x \mapsto mx$ with $m > 0$. Indeed, core zero-handles are selected by the vanishing or positivity of triangle and quadrilateral counts, core one-handles by the vanishing or positivity of their normal face-arc counts, and endpoint-gap incidences by homogeneous comparisons of the corresponding partial sums. Multiplication by $m > 0$ preserves all these decisions. It changes the numerical source-chamber and endpoint-rank addresses, but not the finite handle identifiers or their incidence cycles. The spanning-forest choices and their literal row costs follow.

A simple certificate lists the source side pair of every endpoint-gap two-handle, the ordered pairs

$$(\text{three-handle identifier}, \text{two-handle identifier}),$$

and the surviving source two-handle identifiers. An independent checker reconstructs the side table, checks every selected handle is live, and requires exactly one surviving interior side, with its multiplicity equal to one. It then removes that three-handle and two-handle. At the end it requires that no three-handle survives and that the supplied survivor list is exact. A source side that belonged to an earlier removed three-handle is then exposed; a loop with two live occurrences fails the singleton test. Survivors can be densely renumbered for a downstream checker, while the certificate retains the original source identifier map. Other records, in particular all zero/one identifiers and ports, retain their original indices.

Proposition 2.7 (Combinatorial and bit costs)**.** *Let t be the number of source tetrahedra and let L bound the bit length of the normal coordinates. Given authenticated source incidence and coordinate data, dual graph construction, a breadth-first spanning tree, and trace replay require $O(t)$ combinatorial incidence operations. The added trace and side table contain $O(t \log(t+1))$ bits, independent of normal weight magnitudes. Recomputing the local weights by (1) adds $O(tL)$ elementary bit work. Independent source reconstruction and validation are additional costs.*

Proof. There are at most $4t$ source vertex flags, at most $6t$ local edge occurrences, and at most two core endpoint gaps per global interior edge. The vertex-link flag lists have total length $4t$. Every source edge-link wedge appears in at most two endpoint-gap cycles, so the total number of two-handle zero/one flag occurrences is $O(t)$. Building the side table, adjacency lists and rooted tree, and scanning a supplied trace therefore use $O(t)$ incidence operations. There is one move per interior vertex. All identifiers and owner labels are bounded by $O(t)$, proving the bit size bound. A local edge weight is a sum of a bounded number of L-bit coordinates, and the finite chart scan has $O(t)$ such operations. This proves the bit-work bound. It does not identify the cost of the existing source-preparation routine with this already-prepared graph cost. □

2.4 Choosing the top forest optimally for retained work

A rooted breadth-first tree guarantees completion of the top phase, but different trees delete different two-handle rows. This gives an exact and inexpensive optimization that does not presume that incidence size determines downstream recognizability.

Proposition 2.8 (Spanning trees are exactly the complete deletion sets)**.** *For the source-side cancellation rule used here, the sets of two-handles deleted by complete top-cancellation traces are exactly the spanning trees of G_x.*

Proof. One direction is Theorem 2.4. Conversely, direct each selected edge from the three-handle removed at that move to its other endpoint. That endpoint is $*$ or a vertex removed earlier. Following directed edges strictly decreases deletion time until it reaches $*$. Thus there is no directed or undirected selected cycle, and every nonroot vertex has a selected path to $*$. There are exactly $|V_{\mathrm{int}}|$ selected edges. They form a spanning tree. □

Theorem 2.9 (Minimum retained additive incidence)**.** *Assign a nonnegative cost c_h to each source two-handle h, and keep all costs fixed during the top reduction. A maximum-weight spanning tree of G_x, with edge weight c_h, minimizes*

$$\sum_{h \notin D} c_h$$

among every complete pure top-cancellation trace. Examples include the total number of raw attaching occurrences or the number of distinct nonzero incidences in a row after a fixed zero/one forest contraction. The tree can be computed using $O(t \log(t+1))$ graph comparison operations.

Proof. Unselected two-handle rows do not change, so retained cost equals $\sum_h c_h - \sum_{h \in D} c_h$. Proposition 2.8 says exactly which deletion sets are allowed. Maximizing the second sum is the maximum-weight spanning-tree problem on a multigraph with $O(t)$ edges. Sorting the edges and applying the usual greedy forest algorithm proves the complexity statement. Loops cannot be selected; parallel edges are treated as distinct edges. □

This optimum concerns an additive retained-work objective. It does not assert that the same tree maximizes the number of later cancellations, produces the smallest final certificate, or finds a handlebody whenever one exists. Those are different optimization or recognition questions.

2.5 Combining a dual forest with exact raw cancellation

Fix a zero/one spanning forest and a source-authenticated nonnegative raw incidence matrix $N = (n_{rh})$. Its rows include every source two-handle and may include additional geometrically authenticated two-handles from later bundle constructions; these additional rows are not eligible dual edges. Its columns are the remaining one-handles. A nonempty row set R is a stopping set if

$$\sum_{r \in R} n_{rh} \neq 1 \quad \text{for every column } h.$$

Because the entries are nonnegative integers, these sums are zero or at least two. This is the established unsigned stopping condition, not an algebraic rank or exponent-sum test.

For a rooted tree D, let $N \setminus D$ denote deletion of its eligible source rows. A raw positive pivot at (r, h) requires $n_{rh} = 1$, deletes that row and column, and updates the surviving entries by

$$n'_{ij} = n_{ij} + n_{ih} n_{rj}.$$

The prior complete raw-cancellation theorem [10, 7] says that this operation can remove all rows exactly when no nonempty stopping set exists. Equivalently, the unit-column peeling algorithm then removes all rows. This prior theorem supplies the following cut-relative consequence.

Theorem 2.10 (Tree transversal criterion)**.** *For a fixed authenticated matrix and zero/one forest, the following conditions are equivalent:*

1. *Some complete source two/three forest is followed by a complete sequence of positive raw one/two cancellations.*
2. *Some spanning tree D of G_x satisfies* $\mathrm{Stop}(N \setminus D) = \varnothing$.

3. *Some spanning tree D of G_x meets every nonempty stopping set of N, where meeting means sharing an eligible source row.*

Proof. The possible top deletion sets are exactly the spanning trees by Proposition 2.8. Their effect on raw incidence is literal row deletion. Applying the prior raw theorem proves the equivalence of the first two conditions. A row set of $N \setminus D$ is a stopping set precisely when it is a stopping set of N disjoint from D: its entries and columns have not changed. Therefore no such set remains exactly when D meets every original nonempty stopping set. □

2.6 A native five-tetrahedron example with strict coverage gain

The included fixture `fast/core_dual_fixtures.py`, function `interior_loop_torus()`, gives a small exact example in which forest selection changes the existence of *any* complete positive raw continuation. This is stronger than observing different outcomes from two greedy one/two producers.

Here are its complete face identifications. Tetrahedra and local vertices are numbered from zero. A permutation word $p_0p_1p_2p_3$ means that source vertex i maps to target vertex p_i. The reverse gluing is the inverse permutation. The two unpaired faces are $(0, 3)$ and $(1, 2)$.

one-handle	paired faces	vertex permutation
0	$(0,0) \leftrightarrow (4,1)$	1230
1	$(0,1) \leftrightarrow (2,0)$	1023
2	$(0,2) \leftrightarrow (1,3)$	0132
3	$(1,0) \leftrightarrow (2,3)$	3201
4	$(1,1) \leftrightarrow (3,0)$	1023
5	$(2,1) \leftrightarrow (4,3)$	0321
6	$(2,2) \leftrightarrow (3,3)$	0132
7	$(3,1) \leftrightarrow (4,2)$	0213
8	$(3,2) \leftrightarrow (4,0)$	1302

All nine gluing permutations are odd, so the displayed tetrahedron orientations agree. The source equivalence relations have two global vertices and eight global edges. Three edges lie in the boundary; the remaining five are interior. The boundary consists of two triangles with one vertex and three edges and is a connected torus. The finite vertex-link data are

source vertex	link vertices	link triangles	unpaired link sides	χ(link)
0 (boundary)	10	12	6	1
3 (interior)	6	8	0	2.

The same face equivalence check verifies that edge links are intervals or circles and that there is no reversed edge identification. Thus the vertex links are respectively a disc and a sphere; these checks certify the finite-manifold input contract independently of any historical fixture-generation method.

Use the zero normal coordinate vector. There is no parallelity bundle and the cut is the source manifold itself. The source core has handle counts $(5, 9, 5, 1)$, and its canonical zero/one forest is $T = \{0, 1, 2, 4\}$. In source two-handle order, the complete unsigned one-handle passage lists and the dual side owners are

r	source edge identifier	ordered passage list	dual side owners
0	2	$(2, 4, 6, 1)$	$(*, v)$
1	4	$(2, 3, 5, 0)$	$(*, v)$
2	5	$(1, 5, 8, 7, 0)$	$(*, v)$
3	11	$(4, 7, 8, 6, 3)$	$(v, *)$
4	14	$(6, 7, 5)$	(v, v).

The vertex v denotes the single three-handle. The last dual edge is a genuine loop with two incidences at v and is ineligible for its cancellation. In columns $(3, 5, 6, 7, 8)$, contraction of T leaves

$$N = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 \end{pmatrix}. \tag{2}$$

These lists can be reconstructed simply by walking the five interior source edge links in the face table. The delivered independent source checker performs that reconstruction before either top-forest replay or raw incidence replay.

Proposition 2.11 (A strict source-realized difference between forests)**.** *For the source in* (2)*, the canonical breadth-first top forest leaves a matrix that admits no full positive raw cancellation. The maximum retained-work saving forest leaves a matrix that does admit a complete no-fill cancellation, proving that the source manifold is a solid torus. Their retained raw occurrence counts are respectively twelve and nine.*

Proof. The breadth-first producer encounters the edge of row zero first and deletes that two-handle with v. The four remaining rows have column totals

$$(2, 3, 2, 3, 2).$$

Their entire row set is therefore a stopping set. Stopping persistence excludes every complete sequence of positive raw pivots, including sequences that introduce fill. This failure is not tied to the later producer's pivot order.

The five raw row costs are $(1, 2, 3, 4, 3)$. Eligible edges are rows zero through three; row four is a loop. A maximum-weight rooted tree therefore uniquely chooses row three, with cost four. Delete it with the three-handle. The surviving rows have the following unit-column peeling sequence, written in the original source row and one-handle labels:

$$(1, 3), \quad (2, 8), \quad (4, 5), \quad (0, 6).$$

At each step the named column has total exactly one among the remaining rows. Hence every cancellation is geometric and introduces no fill. After contraction of the four zero/one forest edges, the four one/two cancellations leave one zero-handle, one one-handle, and no two- or three-handles. The result is a genus-one handlebody. Since the normal surface was empty, this is a direct geometric certificate that the source manifold itself is a solid torus, requiring neither an irreducibility assumption nor an external recognition oracle.

The initial raw occurrence total is thirteen. Deleting a row of cost one leaves twelve, while deleting the row of cost four leaves nine. □

The example establishes a strict useful gain for the weighted producer. It does not show that this cost objective always finds a successful tree whenever one exists. Its complete source contract, small face table, integer passage lists, and positive certificate make it reproducible without the tool that was originally used to discover the fixture.

2.7 Forest-independent obstructions and finite search

Corollary 2.12 (Graphic-rank obstruction for every top forest)**.** *Let $R_1, \ldots, R_k$ be pairwise disjoint nonempty stopping sets of N. Let U be the set of eligible source dual edges whose row labels belong to $\bigcup_i R_i$. If*

$$r_{G_x}(U) < k,$$

where $r_{G_x}(U)$ is the graphic-matroid rank of U, then no top forest followed by positive raw cancellation can remove all rows. In particular, a stopping set consisting only of ineligible rows and dual loops is such an obstruction.

Proof. A successful spanning tree must meet every R_i by Theorem 2.10. Since the R_i are disjoint, these meetings require at least k distinct edges of $D \cap U$. But a subset of a spanning tree is a forest, so $|D \cap U| \leq r_{G_x}(U)$. This contradicts the strict inequality. If all eligible edges of one stopping set are loops, its rank is zero. □

This obstruction has a short independently checkable form: list the disjoint row sets, check their unsigned stopping sums from the source matrix, and compute

$$r_{G_x}(U) = |V(G_x)| - c(V(G_x), U),$$

including isolated vertices in the component count. This is an obstruction to the stated compiler family, not a certificate that the core is not a ball or that the knot is nontrivial.

Proposition 2.13 (Raw-peelable row sets need not form a matroid)**.** *For a fixed nonnegative integral incidence matrix, let an independent row set mean one whose induced row submatrix admits complete raw cancellation. These sets form a hereditary independence system, but in general do not satisfy matroid augmentation. A counterexample of minimum possible row and column dimensions is*

$$N = \begin{pmatrix} 1 & 2 \\ 1 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \textit{with rows } a, b, c.$$

Proof. Every row subset of a matrix with no nonempty stopping set also has no nonempty stopping set. The established raw theorem therefore proves heredity.

Set $I = \{a\}$ and $J = \{b, c\}$. The set I is independent by its unit entry in the first column. The set J is independent: its second column has total one and removes b, after which the first column removes c. Thus $|I| = 1 < 2 = |J|$. However, the column totals on $I \cup \{b\}$ are $(2, 3)$, and the totals on $I \cup \{c\}$ are $(2, 2)$. Each is a nonempty stopping set. Neither element of $J \setminus I$ can augment I.

Any hereditary independence system on at most two elements is a matroid, so three rows are necessary. With a single column, a nonempty row set is raw-peelable exactly when it consists of one row with entry one. The other rows are loops of the independence system. This is a rank-one uniform matroid on the unit rows, together with loops, so two columns are necessary. The displayed matrix attains both lower bounds. □

This is a statement about the general unsigned incidence model; it is not an assertion that this smallest matrix is the full incidence of a particular source cut. It rules out an automatic appeal to matroid intersection for the tree-transversal condition. Any additional exchange property for a narrower family of source triangulations would need a separate geometric theorem. In particular, the additive optimum of Theorem 2.9 should not be advertised as an optimum for downstream complete cancellation.

The failure is not caused solely by doubled individual incidences. For the Boolean rows

$$a = (1, 0, 0), \quad b = (1, 1, 0), \quad c = (0, 1, 1), \quad d = (1, 1, 1),$$

take $I = \{c, d\}$ and $J = \{a, b, c\}$. The set I peels d in column one and then c. The set J peels c, b, a in columns three, two, one. But the totals on $I \cup \{a\}$ are $(2, 2, 2)$, and those on $I \cup \{b\}$ are $(2, 3, 2)$. Thus Boolean incidence already violates augmentation. As before, this is a matrix statement, without an additional claim that these rows form a particular source handle core.

Proposition 2.14 (A finite complete forest search)**.** *Let $q = |V_{\mathrm{int}}|$ and $m = |E(G_x)|$. Enumerating all q-element edge subsets, retaining the spanning trees, and applying complete unit-column peeling to $N \setminus D$ decides the equivalent conditions of Theorem 2.10. Its time is bounded by*

$$\binom{m}{q} \operatorname{poly}(t, L, |N|) \leq m^q \operatorname{poly}(t, L, |N|).$$

Here $|N|$ denotes the encoded raw matrix size. For $q = O(\log t)$ and polynomial encoded matrix size this is quasipolynomial in t, apart from the displayed polynomial dependence on coordinate bit length.

Proof. Every complete top trace deletes one of the enumerated trees. The independent geometric theorem permits every such tree, and exact peeling decides full positive cancellation of its row complement. Thus all candidates in the restricted family are considered. The number of candidates and the polynomial tests give the stated bound. □

In general $q = O(t)$, so this is an exponential worst-case search, not a quasipolynomial unknot-recognition algorithm. Even exhaustive success or failure concerns a fixed source chart system, a fixed zero/one forest, and the positive raw model. Changes of disk system, intersection-reducing slides, new normal cuts and arbitrary boundary-pattern transport are outside the search. The most useful initial implementation is therefore the guaranteed polynomial top phase, optionally with the exact additive tree objective, followed by the existing independently certified consumers.

2.8 A proposed alternating optimization of the two forests

The implemented weighted dual-tree choice optimizes the retained literal attaching-curve work after fixing the primal forest. There is a symmetric, currently unimplemented optimization of the primal forest. The following statement concerns two finite graphs and an incidence matrix; its proof requires no additional claim about transporting boundary markings.

Let G_{01} be a finite multigraph with edge set $\mathcal{H}$, and let G_{23} be a connected finite multigraph with distinguished root and edge set $\mathcal{R}$. Loops are allowed in both graphs but cannot lie in a forest. Let $N = (n_{rh})_{r\in\mathcal{R},h\in\mathcal{H}}$ have nonnegative integer entries. In the intended source application, n_{rh} is the number of literal passages of source two-handle r through source one-handle h. Let T span every connected component of G_{01} and let D be a spanning tree of G_{23}. Define

$$C(T, D) = \sum_{r\notin D} \sum_{h\notin T} n_{rh}, \qquad M = \sum_{r,h} n_{rh}.$$

Thus C counts retained source passages after deleting the selected source two-handles and ignoring the primal-tree one-handles. It uses raw occurrences, without cancellation of inverse letters or any prediction of later handle slides.

Proposition 2.15 (Alternating exact forest responses)**.** *For fixed T, a maximum-weight spanning tree of G_{23} with edge weights*

$$w_T(r) = \sum_{h\notin T} n_{rh}$$

minimizes $C(T, D)$. For fixed D, a maximum-weight spanning forest of G_{01}, spanning each of its components, with edge weights

$$v_D(h) = \sum_{r\notin D} n_{rh}$$

minimizes $C(T, D)$ over T.

Starting from any feasible pair, alternate these two exact responses, accepting a changed forest only when it strictly decreases C. Stop after a complete sweep in which neither response gives a strict decrease. There are at most $C(T_0, D_0) \leq M$ accepted updates. At termination, changing either forest alone cannot lower C.

Suppose the graphs have total combinatorial size $O(t)$ and the matrix is provided as literal occurrence lists of total length M. Then this procedure uses

$$O((M+1)(M+t\log(t+2)))$$

word operations and $O(M+t)$ working storage. In particular, for source incidence of size $M = O(t)$, the bound is $O(t^2 \log(t+2))$ after source construction and authentication.

Proof. For fixed T,

$$C(T, D) = \sum_{r \in \mathcal{R}} w_T(r) - \sum_{r \in D} w_T(r).$$

The first term is constant, proving the dual assertion. For fixed D,

$$C(T, D) = \sum_{h \in \mathcal{H}} v_D(h) - \sum_{h \in T} v_D(h),$$

which proves the primal assertion component by component. These formulas include all loop weights in the constant terms; a loop is never silently treated as a cancellable forest edge.

The objective is a nonnegative integer, so every accepted update lowers it by at least one. Every nonterminal full sweep contains an accepted update. There are consequently at most $M+1$ sweeps, including the final sweep. If an optimal response has the current objective value, retaining the current forest is essential to the specified stopping rule: zero-gain changes are not performed. Thus, in the final unchanged sweep, both current forests individually attain their conditional minimum.

One pass through the occurrence lists computes the required weights. Sorting weighted graph edges and running the usual maximum-spanning-forest algorithm costs $O(t \log(t+2))$ word operations per response. There are at most two responses per sweep. All weights lie between zero and M, so $O(\log(M+t+2))$-bit words suffice for weights and identifiers. This proves the complexity and storage bounds. □

Conditional optimality can be strictly weaker than joint optimality even for two-edge graphs. Give each graph two vertices joined by two parallel edges and take

$$N = \begin{pmatrix} 1 & 2 \\ 2 & 0 \end{pmatrix}.$$

Deleting row two and column two leaves cost one. Changing either deletion alone raises the cost to two, so both responses retain this pair. Deleting row one and column one together instead leaves cost zero. This is an abstract incidence example, without a claimed geometric realization.

This is a proposal for a producer heuristic, not a modification of the runtime delivered here. A final chosen dual tree would still require the independent source-side replay, and a chosen primal forest would still need to be recorded in the applicable downstream certificate. The graph objective does not establish transport of an arbitrary boundary pattern through primal contractions. Nor does conditional optimality prove global optimality over pairs, existence of a complete raw cancellation schedule, or completeness of the recognition algorithm. The strict-update bound is polynomial for the stated literal source representation; for an arbitrary matrix of huge binary-encoded multiplicities, a bound proportional to the numerical value M would instead be pseudo-polynomial.

3 Relative bundle quotients and certificates of strict progress

Lackenby's equality criterion for quadratic handle complexity gives an algebraic way to certify that a proposed surface makes strict progress. The useful object is the part of relative second homology that is represented inside the interior parallelity bundle. A surface whose class survives after quotienting out that subspace cannot be one of the homologically bundle-supported surfaces allowed by the equality case. We make this observation quantitative and give a complete exact certificate system for its algebraic premise.

The geometric implication below is a corollary of [4, Proposition 5.9]. The linear alternative is an application of finite-dimensional duality. The contribution here is their explicit combination with a small, independently replayable certificate contract, including the conditions under which the algebraic result can be promoted to a statement about the geometric algorithm.

3.1 The pair map and the geometric hypotheses

Let M be a compact connected oriented PL 3-manifold with boundary pattern P, and let $\mathcal{H}$ be an admissible handle structure in the sense of [4, Definition 5.3]. Let $\mathcal{B}$ be its *actual interior parallelity bundle*, as defined in [4, Definition 5.6], with the embedding in M inherited from those handles. Throughout this section use the conservative pair

$$A = \mathcal{B} \cap \partial M, \qquad j : (\mathcal{B}, A) \longrightarrow (M, \partial M).$$

This is an inclusion of pairs, so it induces a well-defined map on relative homology. Define

$$V_{\mathcal{B}} = \operatorname{im}\Big(H_2(\mathcal{B}, A; \mathbb{Q}) \xrightarrow{j_*} H_2(M, \partial M; \mathbb{Q}) \Big). \tag{3}$$

All chain complexes used to compute this map must come from compatible cellular or simplicial models of the *embedded* pair. An abstract isomorphism between homology groups does not specify this image.

Remark 3.1 (Horizontal boundary is an additional embedding assertion). One can instead use $A_h = \partial_h \mathcal{B}$ if its actual inclusion in ∂M has been established. For the interior parallelity handles, having no 3-handles is a sufficient condition: the horizontal faces are disjoint from other 0-, 1-, and 2-handles and hence lie in the manifold boundary. A general handle structure may have 3-handles, and horizontal faces may meet them. Thus a map $H_2(\mathcal{B}, \partial_h \mathcal{B}) \to H_2(M, \partial M)$ must not be formed solely from an abstract description of the bundle. The choice $A = \mathcal{B} \cap \partial M$ always defines the pair map needed here. When $A_h \subseteq A$, its image contains the image obtained from $(\mathcal{B}, A_h)$, so it can make the obstruction weaker but does not invalidate the sufficient implication. Twisted I-bundles are allowed.

For a 0-handle H, write $I(H)$ for its integer index from [4, Definition 5.1]. The quadratic complexity is

$$c_q(\mathcal{H}) = \sum_{H \in \mathcal{H}^0} \big(\max\{I(H), 0\}\big)^2.$$

The subscript q denotes quadratic complexity. Later the certificate file also has a field named `q`; we denote that positive integer by ν in the mathematics to distinguish the two uses.

Theorem 3.2 (The quotient obstruction forces a positive even raw defect)**.** *Let $\mathcal{H}$, M, P, $\mathcal{B}$, and A be as above. Let S be a compact connected oriented surface properly embedded in M and normal with respect to $\mathcal{H}$. Suppose*

$$[S] \notin V_{\mathcal{B}}.$$

Let $\mathcal{H}^{\mathrm{raw}}$ be the handle structure obtained by cutting along S, before any deletion, handle cancellation, merging, admissibility repair, or bundle replacement. For each original 0-handle H, let $H_{H,1}, \dots, H_{H,r_H}$ be all *its raw child 0-handles, and put $a_{H,i} = I(H_{H,i})$. Then*

$$c_q(\mathcal{H}) - c_q(\mathcal{H}^{\mathrm{raw}}) = 2 \sum_{H \in \mathcal{H}^0} \sum_{1 \le i < j \le r_H} a_{H,i} a_{H,j} \;\ge\; 2. \tag{4}$$

If $\mathcal{H}^{\mathrm{clean}}$ is subsequently obtained by operations independently proved not to increase c_q, then

$$c_q(\mathcal{H}) - c_q(\mathcal{H}^{\mathrm{clean}}) \ge 2$$

as well. The exact identity in (4) concerns the raw cut.

Proof. Since $0 \in V_{\mathcal{B}}$, the hypothesis implies $[S] \neq 0$ in $H_2(M, \partial M; \mathbb{Q})$. Orientability of M and S makes S two-sided. If S separated M, the closure of one side would give a relative 3-chain whose boundary is S, up to orientation, together with a chain in ∂M. This would imply $[S] = 0$. Consequently S is nonseparating, and the hypotheses of [4, Proposition 5.9] hold.

Admissibility implies $I(H) \ge 0$. Index conservation under a standard cut and nonnegativity after a normal cut of an admissible handle structure give

$$\sum_{i=1}^{r_H} a_{H,i} = I(H), \qquad a_{H,i} \ge 0$$

for every H; these are [4, Lemmas 5.2 and 5.5]. Expanding the square gives

$$I(H)^2 - \sum_i a_{H,i}^2 = \left(\sum_i a_{H,i} \right)^2 - \sum_i a_{H,i}^2 = 2 \sum_{i<j} a_{H,i} a_{H,j}.$$

Summing over all original 0-handles proves the exact formula and shows that the raw defect is an even nonnegative integer.

Suppose that the defect were zero. The equality conclusion of [4, Proposition 5.9] would make S homologous in $H_2(M, \partial M)$ to a vertical surface contained in $\mathcal{B}$. The proof of that proposition arranges the representative to be properly embedded in $\mathcal{B}$, disjoint from $\partial_v \mathcal{B}$, and with boundary in ∂M. In particular its boundary lies in $A = \mathcal{B} \cap \partial M$, so it defines a relative class in $H_2(\mathcal{B}, A; \mathbb{Q})$ whose image is $[S]$. This contradicts $[S] \notin V_{\mathcal{B}}$. The even nonnegative defect is therefore at least two.

For the last assertion, write the total defect as

$$\left(c_q(\mathcal{H}) - c_q(\mathcal{H}^{\mathrm{raw}})\right) + \left(c_q(\mathcal{H}^{\mathrm{raw}}) - c_q(\mathcal{H}^{\mathrm{clean}})\right).$$

The first summand is at least two and the second is nonnegative by the separately established property of the cleanup. □

The formula must retain the complete raw child inventory. A parallelity 0-handle with certified index zero contributes zero and may be omitted from a numerical sum after that fact is established. An already reduced inventory may also have lost positive-index pieces or changed the parent relation, so the conservation identity cannot simply be applied to it. There is no asserted parity for an additional cleanup defect. Irreducibility of M and essentiality of P are not hypotheses of Proposition 5.9 or of Theorem 3.2. They must still be supplied if a surrounding cleanup or hierarchy routine requires them. For disconnected M, apply the argument in the component containing S and add the unchanged complexities of the other components.

Corollary 3.3 (A bound for a retained monotone sequence)**.** *Consider a retained forward sequence of handle structures starting at $\mathcal{H}_0$. Suppose every intervening operation is c_q-nonincreasing and N of the cuts satisfy Theorem 3.2. Then*

$$N \le \left\lfloor \frac{c_q(\mathcal{H}_0) - c_q(\mathcal{H}_{\text{last}})}{2} \right\rfloor \le \left\lfloor \frac{c_q(\mathcal{H}_0)}{2} \right\rfloor.$$

Proof. Each of the N certified cuts contributes a drop of at least two. All other contributions are nonnegative, so the drops telescope. □

This corollary counts certified cuts on the retained monotone sequence. It does not count discarded attempts, restored states, or later reconstructions after backtracking. It gives additive progress and therefore does not establish logarithmic hierarchy depth. A proof controlling the complete search must also charge the work on histories that are subsequently discarded.

3.2 A complete finite-chain alternative

We now isolate an algebraic problem that can be checked without trusting a proposed basis of bundle cycles. Let the dimensions of C_i and E_i be c_i and e_i. The supplied rational matrices are

$$d_2 : C_2 \longrightarrow C_1, \quad d_3 : C_3 \longrightarrow C_2, \quad \delta_2 : E_2 \longrightarrow E_1, \quad j_1 : E_1 \longrightarrow C_1, \quad j_2 : E_2 \longrightarrow C_2.$$

The input also includes $s \in C_2$ and must satisfy

$$d_2 d_3 = 0, \qquad d_2 j_2 = j_1 \delta_2, \qquad d_2 s = 0. \tag{5}$$

These equations suffice to define

$$H = \ker d_2 / \operatorname{im} d_3, \qquad V = \left(\operatorname{im} d_3 + j_2 \ker \delta_2\right) / \operatorname{im} d_3 \ \subseteq H.$$

If the matrices arise from authenticated relative chain complexes and the actual inclusion of pairs, then $H = H_2(M, \partial M; \mathbb{Q})$ and $V = V_{\mathcal{B}}$. Without that authentication, they define precisely the displayed algebraic quotient and no asserted manifold.

Theorem 3.4 (Primal support or dual obstruction)**.** *For data satisfying (5), exactly one of the following systems has a rational solution:*

$$\text{(P)} \qquad s = d_3 a + j_2 b, \qquad \delta_2 b = 0, \qquad\qquad a \in C_3, \quad b \in E_2; \tag{6}$$

$$\text{(D)} \qquad y d_3 = 0, \qquad y j_2 = u \delta_2, \qquad y s = 1, \qquad\qquad y \in C_2^*, \quad u \in E_1^*. \tag{7}$$

System (P) *holds exactly when $[s] \in V$; system* (D) *holds exactly when $[s] \notin V$.*

Proof. By the definition of V, membership is equivalent to the existence of $b \in \ker \delta_2$ such that $s - j_2 b \in \operatorname{im} d_3$. This is precisely (P). If (P) and (D) both held, then

$$1 = ys = y d_3 a + y j_2 b = u \delta_2 b = 0,$$

which is impossible.

To prove existence of (D) when (P) fails, take a matrix Z whose columns are a basis of $\ker \delta_2$ and put $W = [\, d_3 \mid j_2 Z \,]$. Then $s \notin \operatorname{col} W$. Define a linear functional on $\operatorname{span}(\operatorname{col} W, s)$ to vanish on $\operatorname{col} W$ and to take value one on s, and extend it to C_2. Call its row vector y. It satisfies $y d_3 = 0$, $ys = 1$, and $y j_2 Z = 0$.

For completeness, a row v annihilates $\ker \delta_2$ if and only if it is in the row space of δ_2. One direction follows by multiplication. For the converse, the rule $\delta_2 b \mapsto vb$ is well-defined on $\operatorname{im} \delta_2$ when v vanishes on its kernel; extending this functional to E_1 gives a row u with $v = u\delta_2$. Apply this fact to $v = y j_2$. It gives $y j_2 = u \delta_2$, completing (D). □

The matrix Z occurs only in this proof. The certificate does not accept a purported cycle basis as trusted input. A basis with missing vectors could make $yj_2Z = 0$ hold while y fails to annihilate the images of other bundle cycles. The identity $yj_2 = u\delta_2$ checks annihilation of *every* bundle cycle directly.

Only the displayed chain fragment is needed for the algebraic calculation. If a full inclusion $j_* : E_* \to C_*$ is supplied, the boundaries in E_2 already map to boundaries in C_2 through $j_2\delta_3 = d_3j_3$; enumerating E_3 is unnecessary for computing the image from all cycles of E_2. This observation does not dispense with authenticating the full geometric map when that interpretation is claimed.

Corollary 3.5 (Integer certificate form)**.** *For integral input matrices and integral s, exactly one of the following types of integer certificate exists:*

$$(\mathrm{P}_{\mathbb{Z}}) \quad \nu s = d_3a + j_2b, \quad \delta_2 b = 0, \qquad \nu \in \mathbb{Z}_{>0}; \tag{8}$$

$$(\mathrm{D}_{\mathbb{Z}}) \quad Yd_3 = 0, \quad Yj_2 = U\delta_2, \quad Ys = \nu, \qquad \nu \in \mathbb{Z}_{>0}. \tag{9}$$

Here a, b, Y, U have integral entries and the same dimensions as above.

Proof. Clear common denominators in a rational certificate, including the coefficient one on the normalized right-hand side. Conversely divide an integer certificate by ν. The resulting rational alternatives are exclusive by Theorem 3.4. Equivalently, if both integer certificates existed with multipliers $\nu_P, \nu_D > 0$, pairing them would give $\nu_P\nu_D = 0$. □

The positive multiplier is essential because the question is over $\mathbb{Q}$. For example, take $C_2 = E_2 = \mathbb{Q}$, let j_2 be multiplication by 2, set $s = 1$, and let all other relevant spaces be zero. The class is supported over $\mathbb{Q}$, with integer certificate $\nu = 2$, $b = 1$. Requiring an integer certificate with $\nu = 1$ would confuse rational image membership with integral divisibility.

The auxiliary row U is also substantive. Take $C_2 = E_1 = E_2 = \mathbb{Q}$, $\delta_2 = [3]$, $j_2 = [2]$, $s = 1$, and $C_1 = C_3 = 0$. Since $\ker \delta_2 = 0$, the class is outside the image. A certificate is

$$Y = [3], \qquad U = [2], \qquad \nu = 3.$$

Its restriction to E_2 is a coboundary rather than the zero row.

3.3 Witness size and exact production

The matrix system used by the reference producer is

$$\mathsf{A}x = \mathsf{f}, \qquad \mathsf{A} = \begin{pmatrix} d_3 & j_2 \\ 0 & \delta_2 \end{pmatrix}, \qquad \mathsf{f} = \begin{pmatrix} s \\ 0 \end{pmatrix}, \qquad x = \begin{pmatrix} a \\ b \end{pmatrix}. \tag{10}$$

Let $R = c_2 + e_1$, $K = c_3 + e_2$, and $r = \operatorname{rank} \mathsf{A}$. Suppose the entries of A and f are integral and have absolute value at most $M \geq 1$. Define

$$D_0 = 1, \qquad D_k = (\sqrt{k}\, M)^k \quad (k \geq 1).$$

Theorem 3.6 (Polynomial-size integer witnesses)**.** *If (10) is consistent, a certificate of type $(\mathrm{P}_{\mathbb{Z}})$ exists with*

$$1 \leq \nu \leq D_r, \qquad |a_i|, |b_i| \leq D_r.$$

If the system is inconsistent, a certificate of type $(\mathrm{D}_{\mathbb{Z}})$ exists with

$$|Y_i|, |U_i| \leq D_r, \qquad 1 \leq \nu \leq D_{r+1}.$$

Consequently every entry of either certificate can be encoded with $O((r+1)(\log(r+2) + \log M))$ bits, and the total certificate length is polynomial in the explicit chain ranks and input-entry bit lengths.

Proof. In the consistent case, choose r independent columns of A and r rows on which they form a nonsingular matrix T. A basic solution sets all other variables to zero. Cramer's rule expresses each remaining variable as the determinant of T with one column replaced by the corresponding entries of f, divided by $\det T$. Because the chosen columns span the column space and the system is consistent, this solution satisfies every row, not only the selected rows. Put $\nu = |\det T|$ and choose the numerator signs accordingly. Hadamard's inequality bounds every determinant just mentioned by D_r. If $r = 0$, consistency forces $\mathsf{f} = 0$, and one can take $\nu = 1$ and all coefficients zero.

In the inconsistent case, $\operatorname{rank}[\,\mathsf{A} \mid \mathsf{f}\,] = r + 1$. Choose r independent columns of A and $r + 1$ rows on which those columns together with f form a nonsingular $(r + 1) \times (r + 1)$ matrix T^+. The cofactors of the last column define an integer row λ, supported on the selected rows. Laplace expansion gives

$$\lambda\mathsf{f} = \det T^+ \neq 0, \qquad \lambda\mathsf{A} = 0.$$

For the second identity, the cofactors annihilate each selected column by the determinant with two equal columns; they then annihilate all columns because the selected ones span $\operatorname{col}\mathsf{A}$. Reverse the sign of λ if necessary and put $\nu = \lambda\mathsf{f} > 0$. Splitting the row according to the two row blocks as $\lambda = (Y, -U)$ gives

$$Yd_3 = 0, \qquad Yj_2 - U\delta_2 = 0, \qquad Ys = \nu.$$

Each cofactor is an $r \times r$ determinant, so its absolute value is at most D_r; the final determinant has absolute value at most D_{r+1}. This also covers $r = 0$, using the determinant of the empty matrix equal to one. Taking binary lengths proves the last assertion. □

These bounds are existence bounds obtained from selected minors. They do not require total unimodularity, saturation of an integral lattice, or a unimodular basis change. Rational input can be reduced to integral input by clearing denominators; the resulting entry lengths remain polynomial in the total rational input length.

Proposition 3.7 (The row-provenance producer is complete)**.** *Over exact rational arithmetic, Gaussian elimination applied to* (10)*, while retaining every elementary row operation on an identity matrix, produces one of the integer certificates in Corollary 3.5. It uses polynomially many arithmetic operations with polynomially bounded operand lengths.*

Proof. Start with the augmented matrix $[\mathsf{A} \mid \mathsf{f}]$ and a provenance matrix $T = I_R$. At every row swap, row rescaling, or row addition, perform the identical operation on T. The maintained invariant is

$$[\,\widetilde{\mathsf{A}} \mid \widetilde{\mathsf{f}}\,] = T[\,\mathsf{A} \mid \mathsf{f}\,].$$

The producer selects pivots only in coefficient columns and eliminates each pivot from all other rows.

If a row has zero coefficients and nonzero right-hand side β, let t be its row of T and set $\lambda = t/\beta$. Then $\lambda\mathsf{A} = 0$ and $\lambda\mathsf{f} = 1$. Writing $\lambda = (y, -u)$ gives $(yd_3, yj_2 - u\delta_2) = (0, 0)$ and $ys = 1$. Clearing the denominators of y, u, and the extra entry 1 produces integer (Y, U, ν) with $\nu > 0$. The minus sign in the lower block is necessary and is the sign used by the reference implementation.

If no inconsistent row occurs, assign zero to every nonpivot variable and the reduced right-hand-side entry to each pivot variable. Reduced echelon form then gives a solution $x = (a, b)$. Clear the denominators of its entries and the extra entry 1. This gives (a, b, ν) satisfying (8), with $\nu > 0$. Dividing all certificate entries by their common positive greatest common divisor preserves the equations. These arguments also apply when any of the spaces has dimension zero. If $R = 0$, the zero vector of length K with $\nu = 1$ is a support certificate.

There are polynomially many row operations on matrices of dimensions bounded by R and $K + R + 1$. With reduced rational arithmetic, the entries arising in elimination, including the provenance entries, are ratios of minors of matrices formed from the input and an identity block. Hadamard bounds their numerators and denominators by a polynomial number of bits. Products, sums, greatest common divisors, and common-denominator clearing consequently have polynomial bit complexity. This proves the claim for the exact algorithm implemented using `Fraction`. □

The size convention in the last two statements counts the declared chain ranks, in addition to matrix-entry bit lengths. It agrees with a cellular input that explicitly lists its generators. It is not a claim of polynomial complexity in bare JSON byte length for a succinct declaration of huge empty matrices: a file can declare K unconstrained generators using $O(\log K)$ digits, whereas the current producer writes a vector of length K.

3.4 Independent replay and its exact scope

The implementation in `research/quotient` separates a rational producer from an integer verifier. The input schema records the five dimensions, all five matrices, and s explicitly; the matrix called δ_2 above is the JSON field `e2`. Dimensions are retained even for empty matrices. The verifier rejects booleans, floating-point values, strings used as integers, inconsistent shapes, unrecognized fields, and failures of (5).

Proposition 3.8 (Soundness of independent integer replay)**.** *For input satisfying* (5)*, the following replay rules are sound:*

1. *accept a support certificate only after checking strict integer types, dimensions, $\nu > 0$, and all identities in* (8)*;*
2. *accept an outside certificate only after checking strict integer types, dimensions, $\nu > 0$, and all identities in* (9)*.*

The first verdict proves $[s] \in V$. The second proves $[s] \notin V$. Replay uses only exact integer products, sums, and equality checks, and need not invoke the producer, Gaussian elimination, or a homology basis computation.

Proof. Dividing the first certificate by ν gives (P), and hence membership. For the second, suppose $[s] \in V$. Then $s = d_3 a + j_2 b$ for rational a, b with $\delta_2 b = 0$. The replayed identities would imply

$$\nu = Ys = Yd_3 a + Yj_2 b = U\delta_2 b = 0,$$

contradicting $\nu > 0$. The operations stated in the proposition are exactly those needed to evaluate the identities. □

The verdict strings are `INSIDE_BUNDLE_IMAGE` and `OUTSIDE_BUNDLE_IMAGE`. For arbitrary input matrices these refer to the algebraic subspace V defined above. In particular, an inside verdict does not assert that a geometric cut is neutral: Lackenby's implication is that equality of complexity forces bundle support, not that every bundle-supported surface realizes equality. An outside verdict becomes a strict geometric progress certificate only after all hypotheses of Theorem 3.2 have been authenticated.

The certificate also records a SHA-256 digest of a canonical JSON serialization of the input. This is useful provenance metadata and rejects ordinary stale-input mistakes. Algebraic soundness does not rest on a collision-resistance assumption: the identities are evaluated against the supplied input itself.

The exact algorithm and the default command-line integer codec must also be distinguished. This artifact uses JSON decimal integers. CPython's default limit on conversion to or from very long decimal strings can reject a mathematically valid large input or output. The documented launch setting `PYTHONINTMAXSTRDIGITS=0` is required for trusted research instances exceeding that limit. A subprocess test exercises a 16384-bit input coefficient, the resulting 16384-bit support multiplier, and separate producer and verifier commands under this setting. The quotient wire format does not use the hexadecimal codec of the separate raw-cut artifact.

3.5 The dual class and component selection

The outside row is not a surface-producing cohomology class of degree one. If the chain complexes come from the indicated pairs, $yd_3 = 0$ says that y is a relative degree-two cocycle, and $yj_2 = u\delta_2$ says that its restriction to the bundle pair is a coboundary. Thus its class belongs to

$$\ker\Big(H^2(M, \partial M; \mathbb{Q}) \longrightarrow H^2(\mathcal{B}, A; \mathbb{Q})\Big). \tag{11}$$

Over $\mathbb{Q}$, universal coefficients and Poincare–Lefschetz duality give the identifications

$$H^2(M, \partial M; \mathbb{Q}) \cong \mathrm{Hom}_{\mathbb{Q}}\big(H_2(M, \partial M; \mathbb{Q}), \mathbb{Q}\big) \cong H_1(M; \mathbb{Q}).$$

Consequently the separating functional is represented by an *absolute one-cycle* z, and its value on $[S]$ is the algebraic intersection of z with S.

By contrast, the *surface class* $\sigma = [S]$ is the Poincare–Lefschetz dual of a class $\alpha \in H^1(M; \mathbb{Q})$. With compatible sign conventions the relevant evaluation is $\alpha(z) = y(\sigma)$. It is α, rather than y, that a cocycle-to-normal-surface construction should realize. There is no canonical identification sending the separating row itself to the required class of $H^1(M; \mathbb{Q})$.

For example, in a compatible triangulation with relative fundamental 3-cycle F, fix a cap-product convention and send an absolute simplicial 1-cocycle c to the relative 2-cycle $s = c \cap F$. Its class is the dual of $[c]$. This gives a chain-level route from an authenticated cocycle to the input of the quotient verifier without expanding an exponentially large normal surface. Authenticating F, the cap product, and the signed intersection convention remains necessary.

Lemma 3.9 (Reuse of a witness on a component)**.** *Suppose integer rows Y, U satisfy $Yd_3 = 0$ and $Yj_2 = U\delta_2$. Let $s_1, \dots, s_m$ be integral relative cycles, and suppose*

$$Y\left(\sum_{i=1}^{m} s_i\right) > 0.$$

Then some i has $Ys_i > 0$. For any such i, the same rows Y, U, with positive multiplier $\nu_i = Ys_i$, are a valid outside certificate for s_i.

Proof. The displayed positive number is the sum of the integers Ys_i, so at least one summand is positive. The two annihilation identities do not depend on the candidate cycle. Substituting s_i and $\nu_i = Ys_i > 0$ therefore gives all of (9). □

Apply the lemma to the oriented components of a properly embedded normal representative. It shows that a future compressed component extractor need only find one component with positive evaluation under a fixed signed scalar functional; computing a complete homology vector for every component is unnecessary for this selection. A normal component with nonzero evaluation has class outside $V_{\mathcal{B}}$ and is therefore nonseparating. A serialized certificate for the selected component must also bind to that component's input digest and use its new positive multiplier; only the annihilating rows are reused. The statement requires the component cycles and their orientations to be authenticated. Unoriented normal coordinates or unsigned component weights do not establish the required evaluations.

3.6 What a polynomial geometric progress step would require

Suppose the quotient $H_2(M, \partial M; \mathbb{Q})/V_{\mathcal{B}}$ is nonzero. At the level of existence, choose a class σ surviving in this quotient, multiply it by a positive integer so that it is integral, and use its dual class in $H^1(M; \mathbb{Z})$. The cocycle construction in [4, Theorem 6.2 and its proof] gives a route to a properly embedded cooriented normal representative. On a triangulation, one normalizes a cocycle to vanish on a maximal tree, places signed intersection points on edges, and joins them by cooriented normal arcs and disks. The resulting total relative class is the prescribed dual class. For patterned handle structures, the cell construction and local normalization require the stated local type bounds; the theorem additionally assumes no 3-handles when $\partial M \neq \varnothing$. This hypothesis must be retained when invoking that version of the construction.

Class-preserving normalization may split the representative. Keeping all components preserves its total relative class; discarding only components with zero relative class is also safe. At least one component survives the quotient. More precisely, first obtain an outside witness for the total class and use Lemma 3.9 to select a component with positive evaluation. Keeping an arbitrary component is not justified: it may lie in $V_{\mathcal{B}}$ even though the total class does not.

Proposition 3.10 (A conditional geometric implementation interface)**.** *Assume the following operations are available with polynomial running time and polynomial output bit budgets in the maintained geometric representation:*

1. *a compiler and verifier for the actual admissible handle structure, its interior parallelity bundle, the embedded pair $(\mathcal{B}, A) \subseteq (M, \partial M)$, and the resulting chain maps;*
2. *a rational cohomology basis and an explicit chain-level Poincare–Lefschetz duality map;*
3. *construction and normalization of a prescribed integral cohomology class as a compressed cooriented normal surface, preserving the total relative homology class and satisfying the relevant handle hypotheses;*
4. *extraction of a connected oriented component with positive evaluation under a supplied signed scalar functional whenever the total evaluation is positive, together with an authenticated relative cycle for that component.*

If $H_2(M, \partial M; \mathbb{Q})/V_{\mathcal{B}}$ is nonzero, these interfaces and the quotient producer and verifier give a polynomial-time construction of a connected oriented normal surface certified to have raw c_q-defect at least two.

Proof. The duals of a rational basis of $H^1(M; \mathbb{Q})$ span $H_2(M, \partial M; \mathbb{Q})$, so at least one lies outside $V_{\mathcal{B}}$. Test the finitely many basis vectors by the exact alternative and choose an outside vector. Clear denominators with a positive multiplier, retaining the correspondingly rescaled positive evaluation. The third interface realizes the resulting integral class and preserves its total value under the dual witness. The fourth interface and Lemma 3.9 select a connected normal component with positive evaluation. The independent verifier accepts the same annihilating rows with this new positive multiplier. The authenticated geometric data then permit Theorem 3.2. The number of basis vectors, the linear algebra, the certificate sizes, and the four assumed operations all have polynomial budgets, proving the conditional running-time statement. □

The present artifact implements the finite-chain producer and replay checker. It does not implement the source-to-chain compiler, the prescribed-class geometric normalizer, or the signed compressed component extractor in this proposition. Weighted orbit methods suggest a route to the last interface, but a method for unoriented component counting alone does not supply its authentication or signed evaluations. The quotient can also vanish; the proposition has no

premise in that case. Resolving such states still requires the bundle-supported reductions and the other hierarchy mechanisms.

4 Source-authenticated implementation and integration

4.1 Placement in the existing terminal operator

The top reduction is placed immediately after the existing compressed parallelity decomposition and before disk-bundle absorption or construction of the relative boundary kernel. This order is mathematically significant: the eligible two-handles are exactly the source endpoint gaps. An additional two-handle introduced by a later bundle operation has no source edge-side authentication and must never be inserted into the dual graph.

The producer is implemented in `core_dual_forest.py`. Its independent checker resides in `core_dual_forest_verify.py`. The producer obtains a deterministic rooted tree. The checker independently reconstructs the oriented side table from the source and validates an ordered cancellation trace. It does not call the forest producer, its planner, or its side-extraction helper. Both routines rely on the established source-preparation contract; independent source authentication is an additional enclosing check.

The public terminal query has the following additional argument:

```
normal_cut_handlebodies(
    triangulation, coordinates,
    record_certificate=True,
    dual_strategy="max_weight",  # default
)
```

The alternatives are `"bfs"` and `"disabled"`. The latter preserves the old conservative three-handle refusal, and is useful for controlled comparisons. The default changes to maximum weight because it has an exact retained-incidence guarantee and a demonstrated strict coverage advantage. There is no assertion that it always finds a successful forest whenever one exists. The three modes take the same downstream code path when the source already has no three-handles.

After a successful top reduction the producer creates a derived core with the original zero/one records, the surviving two-handle records with dense new identifiers, and an empty three-handle list. It updates the four core handle counts. The Euler characteristic is unchanged, so the original cut Euler identity remains valid. The parallelity bundle, its binary multiplicities, its ports, and the ordered boundary words keep their original source identity. The remaining cancellation and bundle census algorithms then operate on this derived core.

4.2 Two certificate layers

The added local certificate has this exact shape:

```
{
  "schema": "normal-core-dual-forest-v1",
  "source_sides": [[-1, 0], ...],
  "cancellations": [[three_handle_id, source_two_handle_id], ...],
  "surviving_two_handles": [source_two_handle_id, ...]
}
```

The integer -1 denotes exposed boundary. In `source_sides`, order and multiplicity are significant. In particular, `[0,0]` is a loop with two incidences, rather than a cancellable singleton. The survivor list is the exact dense-new-to-original-source identifier map. A checker rejects missing, repeated, dead, out-of-range, or multiplicity-invalid choices and requires every three-handle to be

eliminated. The existing exact hexadecimal integer codec is accepted; Booleans are rejected as integer identifiers.

A complete terminal certificate that used top reduction has an enclosing version-six wrapper:

```
{
  "schema": "normal-cut-handlebodies-v6",
  "dual": { ... local source-relative forest proof ... },
  "downstream": { ... existing v1--v5-shaped proof ... }
}
```

The downstream parallelity certificate authenticates the *original* source geometry. Its geometric consumer proof is interpreted on the derived core only after the checker has authenticated that original geometry and replayed the top reduction. This is not a claim that the original source certificate directly asserts the reduced handle inventory.

The verifier performs the following dependency-ordered operations:

1. Validate the exact wrapper fields and allow only one v6 wrapper.
2. Independently replay the original parallelity and ordered-boundary source certificates against the supplied triangulation and coordinates.
3. Reconstruct and replay the dual forest on that authenticated original core; derive the same dense survivor map and new handle counts.
4. Replay the existing core and bundle certificates on the derived geometry, then independently recompute the census and Euler identity.

Discarding the wrapper cannot silently discard the original three-handles: the downstream proof alone is checked against the unreduced source and fails. A fabricated source side table cannot be made authoritative by changing both the table and the proposed cancellation list, because the source geometry and every side owner are reconstructed first.

The forest certificate deliberately does not certify optimality. Optimality is a property of the producer's chosen strategy and objective, while every valid rooted tree is geometrically safe. Accepting alternative valid trees allows later search heuristics or exact restricted search without changing the trusted checker. Exhaustive small-graph tests check the producer's maximum-weight property separately.

4.3 Operation budgets, arithmetic, and reuse

The six upstream orbit discoveries retain their existing shared cycle allowance. The verifier's `max_operations` option retains its meaning as a cap on the existing orbit-proof operations. It is not a total wall-time or bit-operation cap. Cancellation callbacks cover the added finite source, graph, and replay work and propagate cancellation exceptions. A proof requiring no orbit operation can validly pass a zero orbit allowance even though finite geometric work remains; the integration tests distinguish that case from an input with positive orbit work.

The new side table and tree use only $O(t)$ bounded identifiers. Large coordinates appear in the original source validation, not once per normal disc in the graph. The proof of Proposition 2.7 concerns the added work once source charts have been prepared. The implementation currently reconstructs source information in more than one certificate layer. Consequently, it would be inaccurate to advertise the entire terminal query as a single $O(t)$ routine.

There are two defensible caching levels. First, a cache keyed by the triangulation and edge-zero support can reuse a graph tree while source authenticity is checked anew. Second, scalar multiples of a fixed valid coordinate vector preserve the bounded incidence combinatorics and thus the

maximum-weight choice as well. More ambitious reuse between unrelated surfaces needs a key authenticating all the data used to compute costs. Reusing a valid tree and reusing an optimal tree are different promises.

4.4 The independent homology research component

The separate `research/quotient` directory contains a producer and checker for the integer chain-pair alternative. The producer solves the block linear system with exact rational elimination and row provenance. The checker imports no producer and uses strict integer validation, matrix products, and exact equalities. It checks the chain identities and binds the certificate to a canonical input digest. The digest is useful provenance metadata; the displayed algebraic identities establish soundness.

This code is intentionally not wired to the native cut census. There is no source-to-relative-chain compiler in this contribution. Its random tests and small examples are algebraic instances, not authenticated knot exteriors or measured hierarchy speedups. Matrix ranks, including explicit generator counts, are part of the complexity parameter. A JSON object with a binary-encoded huge dimension and no entries is not treated as a compact representation permitting a huge output vector in sublinear time.

The quotient command-line format uses decimal JSON integers. For trusted research inputs beyond Python's default decimal conversion limit, the documented invocation sets `PYTHONINTMAXSTRDIGITS=0`; a 16,384-bit round-trip test verifies that path. This is separate from the native normal coordinate codec, which already uses exact hexadecimal strings for large wire-format integers. Neither implementation substitutes floating-point rank computations for certificate replay.

4.5 What has and has not been formally verified

The article gives mathematical proofs and the code gives independently implemented executable verifiers. The native source geometry remains in the existing trusted Python verification stack. The new theorem proofs have not been translated into a proof assistant, and Python itself is not a small formally verified kernel. Independence here means that replay does not trust or call the search producer; it is a concrete software boundary, not a claim that the entire repository has been mechanically proved correct.

5 Reproducible validation and performance measurements

5.1 Questions, workloads, and measurement boundaries

The experiments answer four concrete questions. Does the new compiler retain all prior positive classifications? Do its additional results replay against the original source and agree with explicit cuts? Can forest choice change the existence of a complete raw continuation? Does the whole compressed query, including replay, remain usable when multiplicities are large? The experiments do not estimate the distribution of arbitrary knots and do not prove a worst-case recognition bound.

The main workload is the exact maintained 106-case corpus from the pinned revision, with SHA-256

`97b4ceed6f068885760d80dea214538634998dd405b77e76926c1f93db03da5c`.

The separate interior-vertex suite was constructed to exercise vertex links, an interior loop, and repeated exact one/four subdivisions. It has 16 fixtures; three no-three-handle controls are added for paired timings. These are separate targeted workloads, not a claim of 122 independent random samples. Some constructions share their source families.

The recorded environment is Python 3.12.14 on Linux x86-64, with Regina 7.4 for the external oracle. Native certificate production and replay need no Regina call. The timing script records source-file hashes, fixture digests, all samples, and the random seed. Timings refer to this machine and software environment; the saved scripts make the workloads reproducible, not their exact wall-clock values.

5.2 The pinned 106-case corpus

Table 1: Native cut classification on the unchanged maintained corpus. Each positive result was independently replayed.

Outcome	Pinned baseline	Dual BFS	Dual maximum weight
Certified complete census	85	92	92
Inconclusive	21	14	14
Source three-handle refusal	10	0	0
Raw cancellation stalled	8	10	10
Raw ball-cancellation obstruction	3	4	4

Both strategies retain every one of the 85 baseline successes. Their statuses and certified histograms agree on all 106 maintained cases. All source three-handles are removed whenever that stage is reached; three of the ten old refusals proceed to later raw obstructions, while seven produce new complete certificates. The seven gains are listed in Table 2. The table describes components of normal cuts; the occurrence of a trefoil or figure-eight source name must not be read as a claim that that knot is unknotted.

Table 2: Newly certified cuts. H_g denotes a genus-g handlebody and kH_g denotes k disjoint copies.

Maintained case identifier	Exact census
`pilot/interior_1_2/0`	$2H_1$
`pilot/interior_2_3/0`	$2H_1$
`pilot/interior_3_5/0`	H_1
`pilot/finite_trefoil_interior/0`	H_2
`interior_1_2/mixed`	$2H_0$
`interior_3_5/mixed`	$2H_0$
`finite_figureEight_interior/mixed`	$H_0 \sqcup 2H_2$

An independent oracle constructed the actual Regina cut, separated every component, and ran `recogniseHandlebody()` on each. The audit capped explicit expansion at 4,096 normal discs per input and imposed a separate 25-second worker deadline. It completed all 99 inputs under that cap, with no worker error or timeout; seven large binary-coordinate inputs were explicitly not attempted. Of the new method's 92 positive certificates, 85 therefore have direct whole-census oracle comparisons, and every one agrees. The other seven have independent native certificate replay.

The oracle's record status `COMPLETE` means that its computation finished; a recorded genus -1 is Regina's non-handlebody result for that component. This differs from the native operator's `COMPLETE` predicate. Among the 14 native inconclusive cases, eight have only handlebody

components in the oracle, and six contain a non-handlebody component. The new positive compiler has not accidentally converted any of those six into a positive census. The eight positive obstructions provide a focused next research workload rather than a reason to weaken the checker.

5.3 Interior vertices, a genuine loop, and strict weighted coverage

On the 16 additional interior-vertex fixtures, the disabled strategy certifies none, BFS certifies 11, and maximum weight certifies 13. Every positive certificate independently replays and matches the actual Regina cut. The three remaining fixtures contain a non-handlebody component. Repeated exact one/four subdivisions reach nine interior vertices, testing sequences in which later cancellations depend on earlier exposure.

The two additional maximum-weight successes use the five-tetrahedron source of Section 2.6, with the empty surface and with its boundary vertex-link disc. Their censuses are respectively H_1 and $H_0 \sqcup H_1$. Retained raw occurrence counts improve from twelve to nine for the empty surface and from eighteen to fifteen for the disc cut.

The empty-cut result has a proof stronger than a software status comparison. The BFS residual's column totals are $(2, 3, 2, 3, 2)$, so stopping persistence rules out every positive raw continuation. Deleting the maximum-weight row permits the explicit geometric sequence already displayed. The artifact also independently enumerates all small positive-pivot choices and all maximal-minor permanents as a diagnostic cross-check. Neither enumeration is used as a substitute for source geometry or for the theorem-level stopping argument.

5.4 Whole-query paired timings

The timing workload consists of the 16 interior fixtures plus three controls without three-handles. Each has seven rounds with randomized method order, seed 20261012. Successful native timings include certificate production and independent source replay. Explicit timings include construction of the actual cut, component separation, and recognition of every component. The script additionally records Regina A/A repeats to expose ordinary timing variation.

For the 11 interior inputs completed by both native strategies, the ratio of maximum-weight median time to BFS median time ranges from 0.9753 to 1.0386, with median 0.9935. This provides no clear evidence of a runtime improvement between the two forest strategies at these sizes. It is consistent with the additional graph optimization being small relative to the full source and bundle verification work. The proved incidence optimum and the strict coverage improvement are stronger conclusions than a speculative speedup inferred from those small timing differences.

Table 3: Selected seven-round median times, in milliseconds. Native times include independent replay. Rows here are completed by both methods.

Source/surface	Discs	Explicit tets	BFS	Max weight	Regina
4-tet source, empty	0	68	2.70	2.79	0.51
4-tet Möbius, 1×	4	91	4.06	4.17	0.91
4-tet Möbius, 8×	32	231	28.71	29.82	2.29
4-tet Möbius, 64×	256	1,351	29.99	29.45	9.38
4-tet Möbius, 512×	2,048	10,311	29.58	29.18	73.64
28-tet subdivided source, empty	0	492	21.28	20.85	9.80

For the 512-fold Möbius vector, the exact result is 257 solid tori and the median paired explicit-to-maximum-weight ratio is 2.497. Regina is faster on most smaller examples, including the empty four-tetrahedron source. Figure 1 shows the measured transition in this one family.

This is evidence that compression can matter to the complete cut query, not an asymptotic fit or a performance claim for arbitrary inputs.

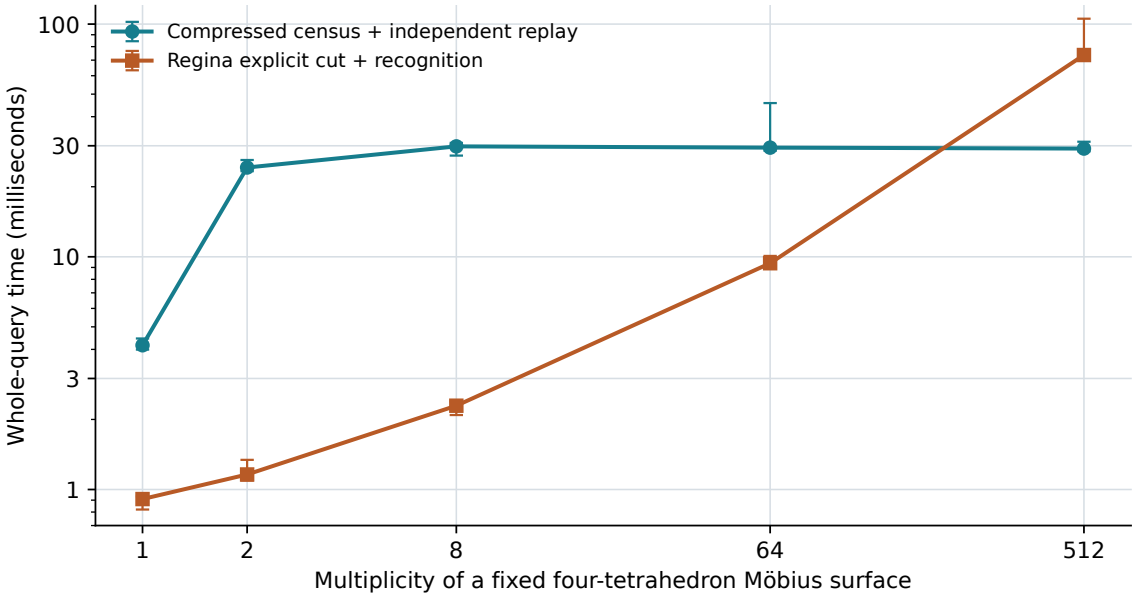


Figure 1: Whole-cut production and verification versus explicit cut and recognition, on one fixed source with increasing Möbius multiplicity. Points are seven-round medians; bars show the observed minimum and maximum. Both axes are logarithmic. Connecting lines only guide the eye.

On a source with three-handles, the disabled path returns early and does not do comparable successful work. Its shorter runtime is an abstention cost and is not used to claim a speedup or slowdown. On the three controls without three-handles, all three strategies take the identical native path. Their measured differences are timing variation, not forest work. The final patch changes the default keyword from BFS to maximum weight after these measurements; every measured call supplied an explicit strategy, so that default change does not change the measured operation.

5.5 Binary capacity and certificate size

The separate capacity run multiplies the four-tetrahedron Möbius vector by 2^{16384}. Both native strategies certify and independently replay the exact census

$$\left(2^{16383}+1\right) H_1.$$

The recorded single-run production-plus-replay totals are 0.1003 seconds for BFS and 0.1088 seconds for maximum weight. Each serialized full certificate is 5,429,503 bytes. No explicit cut of this input was attempted, and there is no paired speedup comparison for it.

The full certificate includes the original source and orbit certificates, which repeat some large integers. Its megabyte size must not be confused with the added top-forest trace, whose combinatorial size is independent of the multiplier. The test establishes that this family is handled without one record per normal disc or component. It does not establish optimal certificate encoding or the same time for every input of comparable coordinate bit length. The multiplier itself has 16,385 bits. An additional integration test cuts the five-tetrahedron loop source by 2^{16384} boundary-link discs and replays the exact census $2^{16384}H_0 \sqcup H_1$.

5.6 Regression, hostile replay, and exact algebra

The full integrated native regression suite passes: 1,696 tests in 377.446 seconds, with no failures or errors. This includes the 12 new dual-forest test methods and six new integration methods. The run used the pinned repository's historical fixtures and dependent research directories, not an incomplete extraction of the runtime folder. Its complete log is included in `research/data/native_regression.log`.

The focused native tests cover repeated interior vertices, both occurrences of a loop, positive-weight loop splitting, reversed dependent orders, alternative valid trees, tetrahedron and local-vertex relabelling, source tampering, forbidden Boolean identifiers, the integer codec, operation allowances, callback propagation, and scalar-certificate reuse. The weighted graph producer is checked against every spanning tree on 128 small multigraph instances. Integration tests disable the relevant producer entry points during complete v6 replay and reject wrapper removal, nested wrappers, altered source geometry, and mismatched coordinates.

The quotient component has 27 passing tests. These include 400 deterministic random algebraic instances, 250 independent SymPy rank comparisons, all 32 combinations of zero/nonzero chain dimensions, rational support with a nonunit denominator, an outside witness with nonunit evaluation, commuting-square failures, malformed matrices, stale input binding, hostile witness mutations, and command-line round trips. The additional 16,384-bit subprocess test checks the documented large-decimal launch configuration. These are checks of the exact algebraic contract, not geometric examples of the future hierarchy interface.

6 From local improvements to a global complexity theorem

6.1 A complete cost ledger

The distinction between normal weight and its encoding is essential but insufficient. For a hierarchy state H_i, let s_i be the full encoded state size: source charts, current pattern, bundle identifications, coordinate bits, and all live compressed-word or interval descriptions. Let a_i count candidate transitions that are generated, including later discarded candidates. A useful global accounting identity has the form

$$T(n) \le T_{\mathrm{input}}(n) + \sum_{i \in \mathcal{V}} \left(T_{\mathrm{state}}(s_i) + \sum_{j=1}^{a_i} T_{\mathrm{transition}}(s_i, j) \right),$$

where $\mathcal{V}$ contains every visited search state, including failed branches. A polynomial bound on one summand does not control either $|\mathcal{V}|$ or the maximum s_i.

The orbit-compression framework of Agol–Hass–Thurston supplies the right kind of primitive for avoiding expansion in multiplicity [1]. Lackenby's surface-curve algorithms likewise illustrate operations charged to triangulation size and logarithmic curve weights [5]. The missing global work is to prove that the particular hierarchy's interfaces stay within those input-size regimes and that sufficiently few states and candidates are inspected. An exponential number of short compressed objects is still too many objects.

Proposition 6.1 (A sufficient search-tree accounting theorem)**.** *Suppose a deterministic recognition procedure on an n-crossing input satisfies all of the following for some constants a, b, c:*

1. *The input conversion takes polynomial time.*
2. *Every visited state and every required transition certificate has encoded size at most n^a.*
3. *At most n^b children are generated at any state, and all local work per child and per state is at most n^c.*

4. *Every root-to-leaf path in the actual explored tree has length at most $d(n)$.*
5. *The branching rule is complete, the leaf predicates are sound, and their combination correctly decides both knot answers.*

Then the total running time is $n^{O(d(n)+1)}$. In particular, $d(n) = O(\log n)$ gives $n^{O(\log n)}$, while $d(n) = O((\log n)^2)$ gives $2^{O((\log n)^3)}$.

Proof. The number of nodes is at most $1 + n^b + \cdots + (n^b)^{d(n)}$. Multiplying by a polynomial upper bound on local work, including child generation and verification, gives the stated bound. The final assertions follow by substituting the two depth estimates and taking logarithms. The correctness hypothesis is listed separately because a fast incomplete search would not prove a recognition theorem. □

The proposition is an accounting tool, not a claim that the hypotheses already hold. In particular, it cannot be applied with the depth of one successful path when the algorithm may search much deeper failed paths. Memoization can improve this bound only after an authenticated equivalence relation on states and a bound on distinct reachable equivalence classes are proved.

6.2 Strict quadratic decrease is weaker than multiplicative decrease

Write $q = c_q(H)$ for Lackenby's quadratic complexity. The homological certificate gives a strict raw drop of at least two under its stated hypotheses. Along a retained forward chain whose other operations are independently q-nonincreasing, there can therefore be at most $\lfloor q_0/2 \rfloor$ such strict cuts. This bound is useful: it excludes indefinite repetition of these progress steps. It is generally linear in the initial potential and gives neither logarithmic depth nor a bound on the number of discarded attempts.

An attempted proof must also distinguish no progress from no available certificate. The homology quotient may vanish, or a candidate may lie inside the bundle image, while a geometric cut still lowers q. Conversely, a nonzero quotient does not itself produce an admissible, efficiently encoded connected normal surface. The quotient test supplies a sufficient progress predicate after geometry has been authenticated.

There is a precise stronger target for a surface constructor. For a raw cut, write $I_H \geq 0$ for each parent zero-handle's integral index and $i_{Ha} \geq 0$ for the indices of all its children, so that $\sum_a i_{Ha} = I_H$. Then $q = \sum_H I_H^2$.

Proposition 6.2 (Balanced index splitting forces multiplicative progress)**.** *Suppose $q > 0$. Let $0 < \alpha \leq 1$ and $0 < \delta < 1$. Assume that a set $\mathcal{U}$ of parent zero-handles satisfies*

$$\sum_{H \in \mathcal{U}} I_H^2 \geq \alpha q, \qquad \max_a i_{Ha} \leq (1-\delta) I_H \quad (H \in \mathcal{U}).$$

Then

$$q(H \setminus\setminus S) \leq (1 - \alpha\delta) q(H).$$

The same upper bound holds after any cleanup whose nonincrease of q has been separately established.

Proof. For $H \in \mathcal{U}$, nonnegativity and index conservation give

$$\sum_a i_{Ha}^2 \leq (\max_a i_{Ha}) \sum_a i_{Ha} \leq (1-\delta) I_H^2.$$

For any other parent, the same argument without the balance assumption gives $\sum_a i_{Ha}^2 \leq I_H^2$. Summing proves a drop of at least $\delta \sum_{H \in \mathcal{U}} I_H^2 \geq \alpha\delta q$. The cleanup assertion is immediate from its separately supplied nonincrease. □

For fixed positive α, δ, a forward nonincreasing chain can have only $O(\log(q_0+1))$ such balance stages while its potential remains positive. More precisely, after k of them the potential is at most $(1-\alpha\delta)^k q_0$; integrality bounds k until the potential vanishes or balance ceases to be available. This does not assert existence of a next balance stage. An index-one parent cannot be split into nonnegative integer children with maximum at most $(1-\delta)I_H$. The residual small-index situation requires a separate geometric treatment, and neutral stages and failed searches need their own bounds.

The inequality identifies the missing construction theorem more sharply than the phrase "find a useful surface." One would need to find a compressed surface or a controlled batch that separates a fixed fraction of the squared-index mass, or else returns a certified alternative region whose handling admits an equally strong amortized estimate. An outside-image witness guarantees some separation, but gives no uniform fraction of the mass.

6.3 A sharper raw-cut accounting potential

The same identity gives a useful local reformulation. Define

$$\ell(H) = \sum_H I_H, \qquad P(H) = \sum_H \binom{I_H}{2} = \frac{q(H) - \ell(H)}{2}.$$

Here P counts unordered pairs of index units lying in the same parent. This interpretation is arithmetic and does not assume those units have canonical geometric representatives.

Proposition 6.3 (Pair-separation accounting)**.** *For a raw normal cut satisfying nonnegative integral child indices and per-parent index conservation,*

$$\ell(H \setminus\setminus S) = \ell(H), \qquad P(H) - P(H \setminus\setminus S) = \sum_H \sum_{a<b} i_{Ha} i_{Hb}.$$

Thus an outside-image cut from the progress theorem lowers P by at least one. Along a sequence consisting only of such raw cuts, the number of strict cuts is at most $P(H_0)$.

Proof. Sum the conservation identities for the first equality. Subtract this equality from the quadratic defect identity and divide by two for the second. Each summand is a nonnegative integer. A positive quadratic defect therefore gives an integral drop of at least one, and the final assertion follows by telescoping the nonnegative potential. □

The qualification "raw" is necessary. Nonincrease of q alone does not imply nonincrease of P if a subsequent operation changes ℓ. For example, as abstract index inventories, four indices equal to one and a single index equal to two both have $q = 4$, but their values of P are zero and one. This arithmetic example is not asserted to be an available geometric cleanup. It shows why a proof of a cleanup rule for q cannot simply be relabelled as a proof for P.

The terminal condition $P = 0$ means that every remaining index is zero or one. It does not mean that the manifold is a ball, that its pattern is essential, or that a knot is unknotted. A useful future strategy could separate the problem of reducing index concentration from the classification of this small-index remainder, provided both stages preserve the required boundary data.

There is an even sharper count for a raw-only sequence. Let $r(H)$ be the number of positive-index zero-handles. Every positive parent has at least one positive child, every zero parent has only zero children, and a strict raw cut has some parent with at least two positive children. Thus each strict cut increases r by at least one. Since $r \le \ell$ and ℓ is conserved, the number of strict raw cuts is at most

$$\ell(H_0) - r(H_0) = \sum_H \max\{I_H - 1, 0\} \le P(H_0).$$

The last inequality follows from $k - 1 \leq \binom{k}{2}$ for every integer $k \geq 1$. The pair potential supplies the exact weighted defect; the positive-child count supplies the sharper elementary number-of-cuts bound. Neither extends through arbitrary cleanup without additional conservation or monotonicity information.

6.4 What the dual-forest theorem contributes to this ledger

The top reduction resolves a specific local obstruction with polynomial encoded work and a coordinate-magnitude-independent additional trace. It does not add a large branching factor: the default uses one optimized tree. Its optional complete restricted search has cost $t^{O(v_3)}\,\mathrm{poly}(t, L)$ for v_3 interior source vertices, and hence is quasi-polynomial in the full encoded source input when $v_3 = O(\log t)$. This is a correct parameterized result for the stated compiler family. No argument in this article makes v_3 logarithmic along an arbitrary full recognition hierarchy or proves that that compiler family is topologically complete.

Removing source three-handles also helps align a normal cut with interfaces that require none. However, it preserves the specified bundle ports, not an arbitrary hierarchy boundary pattern, and it does not automatically establish Lackenby-admissibility. These remaining conditions are precisely where a tempting informal composition would otherwise become invalid.

7 Further research questions and concrete next investigations

The following programme is ordered by the dependencies exposed by the proofs and experiments. Each item names a specific mathematical or software deliverable and the evidence that would justify moving to the next step. The questions are proposals, not claims already established by this article.

7.1 Immediate source-compiler research

1. Joint primal and dual forest selection. The paired-forest objective gives exact coordinate optimizations of retained unsigned incidence. Implement it as an optional producer while keeping the existing independent forest and cancellation checker. Measure successful censuses, retained occurrences, total query time, and certificate size separately. Does alternating selection ever improve certified coverage over the current canonical-primal, maximum-dual choice on actual sources? If it does, find a small explicit face table and a stopping-set proof of strictness, as in the five-tetrahedron example. A mere reduction in the objective would not answer the coverage question.

2. Obstruction-guided dual-tree search. Given a failed residual, extract its maximal stopping core. Every successful tree must meet that core in an eligible source row. Branch only on such edges, reject forced cycles immediately, and use the graphic-rank obstruction to prune incompatible families. Prove that each pruning rule preserves all successful trees. The unrestricted worst-case bound remains $t^{O(v_3)}$; the useful question is whether source geometry bounds the number of relevant stopping cores or supplies a smaller structural parameter. A succinct failed-search proof would certify failure of this compiler family, not nontriviality of a knot.

3. The eight surviving positive corpus obstructions. The explicit-cut audit identifies these current targets:

`pilot/fibonacci_lst_04/0` `pilot/fibonacci_lst_06/0`
`pilot/fibonacci_lst_08/0` `pilot/fibonacci_lst_10/0`
`pilot/lst_4_9/0` `pilot/finite_trefoil/0`
`pilot/finite_figureEight/0` `pachner23_5_8_2/mixed`.

They were already beyond the source-three-handle refusal, so the present top reduction alone cannot resolve them. Determine whether another primal forest suffices; otherwise produce a geometrically authenticated intersection-reducing move, change of disk system, or boundary normalization that destroys the stopping core. The completeness of positive raw peeling means that merely searching more positive pivot orders on the same matrix cannot be the remedy. A successful next contribution should include the smallest actual source obstruction it changes and the new geometric replay rule it needs.

4. Source-realizable exchange structure. The abstract peelable-row system is not a matroid. Are there narrower families of endpoint-gap source matrices with a useful exchange or bounded obstruction property? Conversely, can the minimal three-by-two and Boolean four-by-three counterexamples be realized, with their full row systems, by authenticated finite normal cuts? Either result would sharpen the algorithm-design options. No inference about geometric realizability should be drawn from a presentation matrix alone.

5. Avoiding repeated source preparation and proof duplication. The graph trace is small, but the enclosing verifier currently reconstructs some source data in several layers, and the largest demonstrated full certificate exceeds five megabytes. Design an immutable authenticated source object with explicit consumer contracts. Prove which derived facts can be reused without trusting a producer, and measure complete replay time and bytes. For wire-format deduplication, reference shared integer or source blocks through checked identifiers and retain exact source binding. The target is reduced repeated work, not a weaker authenticity check.

7.2 Geometric realization of homological progress

6. Compile the actual relative bundle inclusion. Construct compatible finite chain complexes for $(\mathcal{B}, \mathcal{B} \cap \partial M)$ and $(M, \partial M)$ from an authenticated admissible patterned handle structure. Produce and verify the actual matrices $d_2, d_3, \delta_2, j_1, j_2$, and prove polynomial bounds on their ranks, entries, and construction time in the maintained encoding. The key issue is the embedded inclusion, including signs, horizontal versus vertical boundary, twisted bundles, and any peripheral identifications. An abstract homology computation or an unsigned incidence matrix is insufficient. Validate the first compiler on small triangulations whose relative chain maps can also be built independently by subdivision.

7. Authenticate the prescribed cohomology-class constructor. Build the chain-level duality map and the normal representative of a chosen integral H^1 class, preserving that class through every normalization. Retain the no-three-handle and local-type hypotheses of the handle version of the construction. Can every intermediate operation be expressed using the project's compressed coordinates and verified pattern charts? A useful first theorem would give a polynomial bit bound for one complete prescribed class construction, including the resulting signed cycle certificate.

8. Extract one component using a signed scalar witness. The component lemma suggests a smaller interface than computing a full homology vector for every component. Given one annihilating row Y and an oriented compressed surface with positive total evaluation, find a

component with Y-evaluation positive and certify its actual oriented relative cycle. Develop the signed weighted-orbit semantics and a source-authenticated component selector. The selector must handle cancellation between positive and negative contributions and cannot rely on unsigned counts alone. Compare its representation size against a full vector-valued component calculation.

9. Understand quotient-zero states. When $H_2(M, \partial M; \mathbb{Q})/V_{\mathcal{B}} = 0$, the proposed predicate has no outside class to select. Characterize those states geometrically under the hierarchy's actual admissibility assumptions. Do they admit a bounded collection of bundle-supported moves, a certified product-region description, or a different decreasing invariant? The task is an alternative construction theorem; absence of an outside witness is not a terminal recognition criterion.

10. Simultaneous cuts and changing bundle images. For disjoint oriented surfaces $S_1, \ldots, S_k$, study the rank of their classes modulo the initial bundle image and its relation to the raw quadratic defect of the simultaneous cut. Is there a lower bound such as twice that rank under appropriate hypotheses, or a small genuine counterexample? The one-surface theorem does not prove such a statement: the actual interior parallelity bundle changes after a cut, so sequential quotients cannot silently be identified. A solution needs compatible relative maps and a theorem controlling their change.

7.3 The global quantitative problem

11. Construct balanced cuts or certified alternatives. Proposition 6.2 gives a concrete sufficient condition for multiplicative quadratic progress. Can one find a surface or a controlled batch that splits a constant fraction of the squared-index mass, or else identify a geometrically structured region with a different amortized simplification rule? In particular, analyze the obstruction from index-zero and index-one pieces instead of assuming balance continues indefinitely. The intended result must include a construction cost and compressed output-size bound, not only existence of a balanced surface.

12. Transport a full boundary pattern through top cancellation. The present theorem is relative to the actual bundle ports. A future hierarchy may carry marks across the exposed faces of selected endpoint gaps. Design an explicit map of those curves and arcs through the cancellation neighbourhood, including corner rounding, order, orientation, and component ownership. Prove bounds on the resulting compressed word size and geometric intersection complexity. Is there a forest objective that minimizes this transport cost as well as raw attaching incidence? Without such a theorem, using the native top reduction as an arbitrary patterned hierarchy move would exceed its proved interface.

13. Count discarded histories and rebuilding work. State the exact search algorithm and assign an amortized charge to every candidate, failed surface, restored pattern, and rebuilt source chart. Can a multisurface batch retain enough geometric and homological information to avoid reconstructing a long prefix after one branch fails? The desired bound must apply to all visited histories. A logarithmic bound on one retained successful chain and a polynomial-size certificate for that chain are not enough.

14. Bound the complete maintained representation. Prove polynomial bounds in the original crossing number for every live chart count, compressed word grammar, interval-family list, relative chain rank, matrix entry, and coordinate bit length. Then verify that every consumer is polynomial in that *same* representation. Local type bounds alone do not limit the number of

repeated local pieces, and binary encoding alone does not limit the number of distinct binary records. This accounting should be machine-readable enough that experiments can log every parameter appearing in the theorem.

15. Formulate and prove the exact final asymptotic statement. Choose whether the theorem targets $n^{O(\log n)}$ or the weaker $2^{O((\log n)^3)}$ formulation, and specify all hypotheses, input conversions, arithmetic conventions, and constants hidden by the notation. Establish completeness on arbitrary knot diagrams and correctness of both answers. Only then combine the encoded-state, branching, and explored-depth bounds using Proposition 6.1. The local results in this article provide useful components for that proof, but do not replace it.

7.4 Recommended order of work

The most direct engineering continuation is the joint-forest producer and the eight positive stopping obstructions, with the existing checker kept as the acceptance boundary. In parallel, the most consequential new geometric interface is the authenticated relative chain map, followed by the signed component selector. Those steps make the homological predicate available to real hierarchy states. Balanced construction, pattern transport, and a charge for discarded histories are then the central mathematical obstacles to the global bound. This order keeps empirical coverage gains connected to explicit lemmas and avoids treating a collection of fast local routines as an already complete recognition algorithm.

A Artifact contents and reproduction

The accompanying archive contains the complete LaTeX source, bibliography, vector and raster figures, compiled article, integration patch, source overlay, a compact frozen runtime, independent algebraic code, source fixtures, saved certificates, raw experimental records, and validation logs. The top-level `README.md` is the practical entry point. `SHA256SUMS` authenticates the packaged files against accidental changes; `provenance/source_pin.json` records the baseline revision and workload identifiers.

A.1 Build and focused checks

From the extracted archive, the principal commands are:

```
python reproduce.py --quick
cd article
latexmk -pdf -interaction=nonstopmode -halt-on-error \
  dual_forests_and_homological_progress.tex
```

The quick command checks packaged hashes, runs the 18 focused native test methods and 27 algebraic test methods, and replays saved census packets. It uses Python's standard library for native production and verification; SymPy enables the additional independent algebraic rank comparison. Regina is required only for explicit-cut audits and the paired timing script. The figure script uses Matplotlib. Full command details and the recorded environment are in the README.

The main corpus can be rerun using the frozen runtime with `PYTHONPATH=runtime/fast` and `research/experiments/audit_native.py`; explicit comparisons use `audit_regina.py`. The baseline audit was run on the untouched pinned implementation. It can be regenerated in a checkout of that revision, while the delivered `disabled` mode reproduces its three-handle refusal for diagnostic

Path	Purpose
`article/`	Main TeX file, included sections, bibliography, figures, and the compiled PDF.
`integration/`	Patch against the pinned ProveIt revision and an equivalent nine-file source overlay. Apply one method, not both.
`runtime/fast/`	Frozen runtime and focused fixtures/tests for standalone reproduction of this contribution. It is not a replacement for the full repository and its historical regression data.
`research/experiments/`	Corpus audit, explicit Regina oracle, paired timings, exact matrix witness, figure generation, and summary scripts.
`research/data/`	Original 106-case workload; baseline, BFS, weighted, and explicit-cut JSON; paired timing samples; test logs and machine-readable summaries.
`research/certificates/`	Saved source-bound census certificates, including all seven newly completed maintained cases under both trees.
`research/quotient/`	Standalone exact chain-pair producer, independent verifier, examples, tests, and interface documentation.
`provenance/`	Pinned tree manifests, source identifiers, artifact claim ledger, and integration validation.

comparisons. The paired benchmark supplies every strategy explicitly and records its code hashes.

A.2 Repository integration

Apply `integration/dual_forest.patch` in a complete ProveIt checkout at the pinned revision. It adds the two runtime modules, two focused test modules, three source/experiment helpers, and modifies the producer and independent verifier of the normal-cut handlebody census. The patch adds no third-party dependency to native recognition.

The full regression command, run from `Topology/UnknotRecognition/fast`, is

```
python -m unittest discover -s tests -v
```

Some existing tests require the repository's `reports/` and `synthesis/` data and an older Git source object. Those dependencies are part of the full project, rather than duplicated into this focused archive. The archived successful run used them. A reduced runtime-only extraction should use the quick focused checks instead of treating missing historical research files as algorithm failures.

The integration validation checks that the patch applies to the exact baseline and produces the same changed files as the supplied overlay. The article and algebraic research component can be placed in the repository's preferred incoming or proposal directory independently of applying the runtime patch.

A.3 Independent review and reproducibility limits

The source-relative theorem, loop multiplicities, relative-port statement, native strict example, and v6 integration were reviewed separately from their producer implementation. The homology alternative, duality degrees, chain-map hypotheses, exact defect, and global accounting statements also received a separate mathematical review. These reviews motivated explicit limits in the final text, particularly the raw/cleanup distinction, arbitrary-pattern transport, and the difference between an additive potential drop and a global search bound.

This is a research contribution supplied for further mathematical and code review. Its executable verifiers, exact examples, proofs, and logs make specific claims inspectable. They are not evidence that the unimplemented geometric interfaces already exist, and the selected timing family cannot establish a universal asymptotic complexity class.

B Claim ledger

The following ledger records the logical status of the main statements. "Proved" refers to the mathematical statement in the cited section; "implemented" refers to the specific delivered code and contract.

Claim	Status	Scope and evidence
Ordinary empty-pattern source two/three cancellation	Prior repository result	Credited to the incoming genus-budget contribution; not asserted as new here.
Complete cut-relative source top reduction	Proved and implemented	Theorem 2.4; authenticated source normal cuts, including empty and one-sided surfaces; relative to actual bundle ports.
Top trace independent of normal magnitudes	Proved and tested	Proposition 2.7; original source validation still depends on coordinate bits.
Same-zero-support trace reuse	Proved; tests included	Proposition 2.6; requires source reauthentication; arbitrary same-support costs need not remain optimal.
Minimum retained additive cost	Proved and implemented	Theorem 2.9; fixed primal forest and fixed additive row costs; no general recognition optimum.
Strict native forest-choice advantage	Proved, implemented, explicit-cut checked	Proposition 2.11; complete five-tetrahedron source data and a persistent stopping obstruction.
Raw stopping-set completeness	Prior repository result	Used to characterize the effect of a changed top forest; not a new peeling theorem.
Tree-transversal and rank obstruction	Proved consequences	Theorem 2.10 and Corollary 2.12; failure is restricted to this compiler family.
Peelable-row matroid exchange	Refuted in general	Proposition 2.13; minimal and Boolean abstract counterexamples; no unsupported source realization.
Restricted exhaustive forest search	Proved; not added to the default	Proposition 2.14; $t^{O(v_3)}\,\mathrm{poly}(t, L)$ for polynomial-size source incidence.
Alternating paired-forest optimization	Proved proposal; unimplemented	Proposition 2.15; polynomial for literal bounded source incidence; conditional minima need not be joint minima.
Rational bundle-image certificate alternative	Proved and implemented algebraically	Theorem 3.4; integer replay and polynomial witness bounds; geometry is an additional contract.

Claim	Status	Scope and evidence
Outside-image strict progress	Proved corollary with explicit hypotheses	Theorem 3.2; actual admissible hierarchy and interior bundle required; exact formula before cleanup.
Reuse of a separating row for one component	Proved	Lemma 3.9; oriented signed component data, new evaluation and input binding required.
Polynomial geometric progress constructor	Conditional interface	Proposition 3.10; source chain compiler, prescribed-class normalization, and signed component extraction remain open.
Balanced multiplicative potential decrease	Proved sufficient criterion	Proposition 6.2; no existence claim for a balanced cut, no bound on neutral or discarded work.
Pair-separation potential and positive-child count	Proved raw-cut identities/bounds	Proposition 6.3 and following argument; arbitrary cleanup needs additional hypotheses.
85 to 92 maintained-corpus successes	Reproduced experiment	Unchanged pinned 106-case corpus; all old successes preserved, positive certificates independently replayed.
No false positive in explicit comparisons	Reproduced experiment	85 directly comparable native positives agree; 99 explicit computations complete; seven large cases outside expansion cap.
Native binary-capacity result	Reproduced single workload	2^{16384} scaling; exact $2^{16383}+1$-component census; capacity, not an asymptotic fit.
Full quasi-polynomial unknot recognition	Not proved by this contribution	Global encoded-size, construction, completeness, branching, explored-depth, and pattern-transport obligations remain in Section 7.

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[10] ProveIt research contributions. Stopping sets, saturated bundles, and exact raw cancellation. https://github.com/VladimirReshetnikov/ProveIt/tree/8cb7d9d25ac8cb64a9d147304e3441e98e94c7a4/docs/incoming, 2026. Incoming package ProveIt_Unknot_Stopping_Cores_2026-10-12.zip; especially the stopping-set and column-peeling theorems.

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